

## 1 Overview

The aim of this note is to give a brief introduction to Moishezon variety and Moishezon morphism. The major references for this note are [Kol22], [Fuj83], and [Uen75].

Why study Moishezon morphisms? First, Moishezon spaces have more functorial behavior (compared with projective varieties), as we will see in Section 2. Secondly, from almost any projective variety we can construct a Moishezon space via bimeromorphic modification, making Moishezon spaces versatile in birational geometry. Thirdly, by Artin's fundamental theorem, the category of Moishezon spaces appears naturally in moduli theory. Another compelling reason to consider the Moishezon category is that it allows cut-and-paste operations similar to those we can perform in topology.

This series of talks is organized as follows:

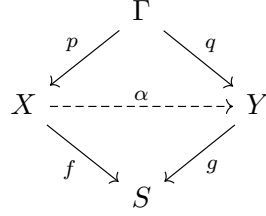
- Lec 1. Basic knowledge about Moishezon spaces and Moishezon morphisms,
- Lec 2. Fiberwise bimeromorphic problems.
- Lec 3. General type locus, Moishezon locus, and projective locus.
- Lec 4. Projectivity criteria and behavior of projective locus.
- Lec 5. Rational curves on Moishezon spaces.

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## 2 Moishezon spaces

**Definition 2.1** (Meromorphic  $S$ -map). Let  $X, Y$  be reduced complex spaces. We call the  $S$ -map a *meromorphic  $S$ -map* if



the natural projection associated to the graph  $p : \Gamma \rightarrow X$  is a proper bimeromorphic morphism. Moreover, if the natural projection  $q : \Gamma \rightarrow Y$  is also a proper bimeromorphic morphism, then we call  $\alpha$  a *proper bimeromorphic  $S$ -map*.

**Remark 2.2** (Comparison between meromorphic map and  $S$ -meromorphic map). By definition,

$$X \times_S Y \hookrightarrow X \times Y$$

is an inclusion. Therefore, it is easy to see that

$$S\text{-meromorphic map} \implies \text{meromorphic map}.$$

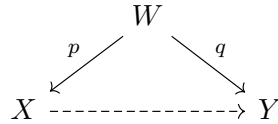
Conversely, the graph of a meromorphic map  $\Gamma \subset X \times Y$  need not be contained in  $X \times_S Y$ , so that a meromorphic map need not be a  $S$ -meromorphic map.

**Remark 2.3** (Comparison between  $S$ -meromorphic map and fiberwise meromorphic map). Note that a  $S$ -bimeromorphic map does not need to be a fiberwise bimeromorphic map, since the restriction of a bimeromorphic map on the subvariety (the fiber) need not be a bimeromorphic map. We will discuss more about the fiberwise bimeromorphic map in the Note-2.

**Definition 2.4** (Moishezon space, first definition). A proper, irreducible, reduced analytic space  $X$  is Moishezon if it is bimeromorphic to a projective variety  $X^{\mathbb{P}}$ .

**Remark 2.5.** The following proposition tells us when the meromorphic map is an actual morphism, using the rigidity lemma.

Let  $f : X \dashrightarrow Y$  be a bimeromorphic map with the resolution of indeterminacy.



then if any  $C \subset W$   $p$ -exceptional is also  $q$ -exceptional. Then the birational map is also a morphism.

**Definition 2.6** (Moishezon space, second definition). A proper, irreducible, reduced analytic space  $X$  is Moishezon if

$$a(X) := \text{tr deg}_{\mathbb{C}} M(X) = \dim(X)$$

that is, it has  $\dim X$  number of algebraic dependent meromorphic function.

**Definition 2.7** (Moishezon space, third definition). A proper irreducible, reduced analytic space  $X$  is Moishezon if it carries a big rank 1 reflexive sheaf  $\mathcal{F}$ . Here the big rank 1 reflexive sheaf means that the induced Kodaira map  $g : X \dashrightarrow \mathbb{P}(H^0(X, \mathcal{F}))$  is bimeromorphic onto its image.

**Proposition 2.8.** The three different definitions for Moishezon spaces above are equivalent.

*Proof.* see e.g. [Uen75]. □

The first important property for Moishezon space is that it locally looks like a quasi-projective scheme up to an étale cover.

**Proposition 2.9** ([Kol22, Proposition 8.2]). Let  $X$  be a Moishezon space. For every  $x \in X$  there is a pointed quasi-projective scheme  $(x', X')$  and an étale morphism  $(x', X') \rightarrow (x, X)$ .

*Proof.* It's quite difficult; for the sake of time, we omit it here. For the curious reader, please refer to [Art70]. □

**Lemma 2.10** (Existence of Galois closure). Let  $\pi : X' \rightarrow X$  be a finite covering between normal analytic varieties. Then there exists a finite Galois covering  $\varphi : X'' \rightarrow X$  from a normal analytic variety  $X''$  which factors through  $\pi$  which is universal in the following sense:

$$\begin{array}{ccc} & X' & \\ \nearrow & & \searrow \pi \\ X'' & \xrightarrow{\varphi} & X \end{array}$$

For any finite Galois covering  $\psi : Y \rightarrow X$  from a normal analytic variety which factors through  $\pi$ , there exists uniquely a Galois covering  $Y \rightarrow X''$  over  $X'$ .

Using the existence of Galois closure, we can write a normal Moishezon space globally as a quotient of a proper variety by a finite group.

**Proposition 2.11** ([Kol22, Proposition 8.3]). Let  $X$  be a Moishezon variety. If  $X$  is normal, then there is a proper variety  $Y$  and a finite group  $G$  that acts on  $Y$  such that  $X \cong Y/G$ . (Note that in general  $Y$  can not be chosen projective.)

*Proof.* First, by Proposition 2.11, there exists some étale cover of  $X$  (indeed, since the étale morphism is finite, we can find an open cover of  $X$  by the étale morphism). Since  $X$  is proper, we can find some finite cover of it. Now by the previous lemma we can take the Galois closure of the finite étale cover  $X_i \rightarrow X$ . We then apply the universal property of the Galois closure, thus it is possible to patch the collection of Galois closures  $\{X_i \rightarrow X\}$  together in the Zariski topology via the gluing lemma (see e.g. Hartshorne Exercise II 2.12.), and therefore we can get a finite covering of  $X$ ,  $Y \rightarrow X$  and thus  $X \simeq Y/G$ . □

Artin [Art70] proved the following theorem, demonstrating the importance of the category of Moishezon spaces in moduli theory.

**Proposition 2.12** ([Art70, Theorem 7.3]). There is a natural functor

$$\text{an} : (\text{algebraic space of finite type over } \mathbb{C}) \rightarrow (\text{complex spaces})$$

extending the functor on the category (schemes of finite type  $/\mathbb{C}$ ). This functor induces an equivalence of categories

$$(\text{complex algebraic schemes of finite type}/\mathbb{C}) \rightarrow (\text{Moishezon spaces}).$$

In other words, every Moishezon space is in a unique way an algebraic space.

We next prove that a Kähler Moishezon space with 1-rational singularity is a projective variety. Before proving the theorem, let us first state two results that will be used in the proof.

**Lemma 2.13.** Let  $X$  be a compact Moishezon space with 1-rational singularity, that is,  $X$  is normal and has a resolution  $\pi : Y \rightarrow X$  such that  $R^1\pi_*\mathcal{O}_Y = 0$ . Then an analytic homology class  $b \in A_2(X, \mathbb{Q})$  is zero if it is numerically equivalent to 0. In particular,

$$A_2(X, \mathbb{Q}) = N_1(X)_{\mathbb{Q}} \subset H^2(X, \mathbb{Q}).$$

**Lemma 2.14** (Nakai-Moishezon criterion for  $\mathbb{Q}$ -line bundles over Kähler Moishezon space). Let  $X$  be a Kähler Moishezon space with a Kähler form  $\omega$ . Assume that an element  $L \in \text{Pic}(X)_{\mathbb{Q}}$  satisfies the equality for any curve  $C \subset X$  :

$$(C.L) = \int_C \omega.$$

Then  $L$  is ample.

**Proposition 2.15** ([Nam02]). Let  $X$  be a Moishezon space with 1-rational singularity. If  $X$  is Kähler, then  $X$  is projective.

*Proof of the Proposition 2.15.* Since the numerical equivalence and the homological equivalence coincide for (analytic) 1-cycles by Lemma 2.13, we have a natural map

$$\alpha : N^1(X)_{\mathbb{Q}} \rightarrow (A_2(X, \mathbb{Q}))^*, \quad d \mapsto (- \cdot d),$$

and  $\alpha$  is an isomorphism (by duality of  $N^1(X)_{\mathbb{Q}}$  and  $N^1(X)_{\mathbb{Q}}$ ).

Note that  $\omega \in H^2(X, \mathbb{R})$  Kähler form as an element of  $(A_2(X, \mathbb{R}))^*$ . By simply defining

$$\alpha_{\omega} : A_2(X, \mathbb{R}) \rightarrow \mathbb{R}, \quad C \mapsto \omega \cdot C = \int_C \omega.$$

Since  $\alpha_{\mathbb{R}}$  is surjective, there is an element  $d \in N^1(X)_{\mathbb{R}}$  such that

$$(C \cdot d) = \int_C \omega,$$

for every curve  $C \subset X$ .

We then approximate  $d \in N^1(X)_{\mathbb{R}}$  by a convergent sequence  $\{d_m\}$  of rational elements  $d_m \in N^1(X)_{\mathbb{Q}}$ .

Let us fix the basis  $b_1, \dots, b_l$  of the vector space  $N^1(X)_{\mathbb{Q}}$ . Each  $b_i$  is represented by an element  $B_i \in \text{Pic}(X)_{\mathbb{Q}}$  via the quotient

$$\text{Pic}(X)_{\mathbb{Q}} \rightarrow N^1(X)_{\mathbb{Q}} = \text{Pic}(X)_{\mathbb{Q}} / \equiv, \quad B_i \mapsto b_i,$$

Now  $d$  (resp.  $d_m$ ) is represented by an element in  $\text{Pic}(X)_{\mathbb{R}}$  (resp.  $\text{Pic}(X)_{\mathbb{Q}}$ )

$$D := \sum x_i B_i,$$

(resp.  $D_m := \sum x_i^{(m)} B_i$ ) such that  $\lim x_i^{(m)} = x_i$ . Put  $E_m := D_m - D$ . Then there are  $d$  closed  $(1,1)$ -forms  $\alpha_m$  corresponding to  $E_m$  such that  $\{\alpha_m\}$  uniformly converge to 0.

If  $m$  is chosen sufficiently large, then  $\omega_m := \omega + \alpha_m$  is a Kähler form. Since

$$(C.D_m) = \int_C \omega_m > 0,$$

for every curve  $C \subset X$ . We see that  $D_m$  is ample by Lemma 2.14 (Note that we have  $D_m$  being a  $\mathbb{Q}$ -divisor, so that it's possible to apply the Nakai-Moishezon criterion). In particular,  $X$  is projective.  $\square$

**Remark 2.16.** There exist some Kähler Moishezon spaces with bad singularity that are not projective. (As we shall see in the last section).

**Proposition 2.17** ([Kol22, Proposition 8]).

- (1) Let  $X$  be a Moishezon space, if  $Z \rightarrow X$  be finite then  $Z$  is Moishezon.
- (2) Let  $X$  be a Moishezon space, and  $f : X \rightarrow Y$  be a surjective morphism of complex varieties. Then  $Y$  is also Moishezon.
- (3) Let  $X$  be a Moishezon space, assume that  $Z \subset X$  is Moishezon, then

*Proof of (1).* By definition

$$\text{trdeg}_{\mathbb{C}} K(X) = \dim X,$$

and if  $Z$  is finite map then

$$K(X) \hookrightarrow K(Z),$$

is a finite field extension. Therefore by additive property for a tower of field extensions, we have

$$\text{trdeg}_{\mathbb{C}}(K(Z)) = \text{trdeg}_{\mathbb{C}}(K(X)) + \text{trdeg}_{K(X)} K(Z) = \text{trdeg}_{\mathbb{C}}(K(X)).$$

$\square$

*Proof of (2).* It will be generalized in to the relative version, see 3.14.  $\square$

*Proof of (3).* Consider the following pull back diagram.

$$\begin{array}{ccc} Z^p = f^{-1}(Z) & \longrightarrow & X^p \\ f_Z \downarrow & & \downarrow f \\ Z & \hookrightarrow & X \end{array}$$

Clearly  $Z^p$  is projective (as subvariety of  $X^p$ ), and  $f_Z$  is surjective (by definition of  $Z^p$ ). Therefore, by (2), we know that  $Z$  is again Moishezon.  $\square$

The following proposition shows that the Moishezon manifolds admit strong Hodge decomposition.

**Proposition 2.18.** If  $X$  is a Moishezon manifold, then the Hodge decomposition holds, indeed a Moishezon manifold admits strong Hodge decomposition.

Before proving the theorem, let us first define what a strong Hodge decomposition is. We say that a compact manifold admits a *strong Hodge decomposition* if the natural maps

$$H_{\text{BC}}^{p,q}(X, \mathbb{C}) \longrightarrow H^{p,q}(X, \mathbb{C}), [\alpha^{p,q}]_{\text{BC}} \mapsto [\alpha^{p,q}]_{\bar{\partial}} \quad \bigoplus_{p+q=k} H_{\text{BC}}^{p,q}(X, \mathbb{C}) \longrightarrow H^k(X, \mathbb{C}), \quad \sum [\alpha^{p,q}] \mapsto \sum \alpha^{p,q},$$

are isomorphisms.

**Remark 2.19.** As a direct consequence, we see that a Moishezon manifold admits *the Du Bois property*, that is

$$H^i(X, \mathbb{C}) \rightarrow H^i(X, \mathcal{O}_X),$$

is surjective for all  $i \geq 0$ . (which will be used in the third note).

*Proof.* The idea of the proof comes from [Dem97, Proposition (12.3)]. We first take the projective modification

$$\mu : \tilde{X} \rightarrow X,$$

such that  $X'$  is a projective manifold. And therefore  $X'$  admits a strong Hodge decomposition. On the other hand

We first observe that  $\mu_* \mu^* \beta = \beta$  for every smooth form  $\beta$  on  $Y$ . In fact, this property is equivalent to the equality

$$\int_Y (\mu_* \mu^* \beta) \wedge \alpha = \int_X \mu^* (\beta \wedge \alpha) = \int_Y \beta \wedge \alpha.$$

for every smooth form  $\alpha$  on  $Y$ , and this equality is clear because  $\mu$  is a biholomorphism outside sets of Lebesgue measure 0 (which holds in general for a proper surjective bimeromorphic map).

Consequently, the induced cohomology morphism  $\mu_*$  is surjective and  $\mu^*$  is injective (but these maps need not be isomorphisms).

$$\begin{array}{ccc} H_{\text{BC}}^{p,q}(\tilde{X}, \mathbb{C}) \longrightarrow H^{p,q}(\tilde{X}, \mathbb{C}), & \bigoplus_{p+q=k} H_{\text{BC}}^{p,q}(\tilde{X}, \mathbb{C}) \longrightarrow H^k(\tilde{X}, \mathbb{C}) \\ \mu_* \downarrow \uparrow \mu^* & \mu_* \downarrow \uparrow \mu^* & \mu_* \downarrow \uparrow \mu^* & \mu_* \downarrow \uparrow \mu^* \\ H_{\text{BC}}^{p,q}(X, \mathbb{C}) \longrightarrow H^{p,q}(X, \mathbb{C}), & \bigoplus_{p+q=k} H_{\text{BC}}^{p,q}(X, \mathbb{C}) \longrightarrow H^k(X, \mathbb{C}) \end{array}$$

Now, we have commutative diagrams with either upward or downward vertical arrows. Hence the surjectivity or injectivity of the top horizontal arrows implies that of the bottom horizontal arrows.

□

We next introduce Campana's Moishezon criterion. The proof uses the core reduction he introduced.

**Proposition 2.20** ([Cam81, Corollaire on p. 212]). Let  $X$  be a compact complex variety in the Fujiki class  $\mathcal{C}$ . Then  $X$  is Moishezon if and only if  $X$  is algebraically connected.

As an immediate consequence.

**Corollary 2.21.** A compact Kahler manifold is projective iff it's algebraically connected.

**Proposition 2.22.** Let  $f : X \rightarrow B$  be a fibration over an algebraically connected variety (e.g. a projective curve). Assume that  $X$  is in the Fujiki class  $\mathcal{C}$  and the general fiber of  $f$  is algebraically connected, then  $X$  is Moishezon if and only if  $f$  has a multi-section.

*Proof.* The proof is clear, since admitting a multi-section implies the algebraic connectedness of  $X$ .  $\square$

**Remark 2.23.** For readers interested in the applications of the algebraic connectedness criterion, I recommend the paper by [Lin23]. He tries to address the following question.

**Question 2.24** (Oguiso–Peternell problem, [Lin23, Problem 1.2]). Let  $X$  be a compact Kähler manifold of dimension  $n$  such that  $\text{Int}(\text{Psef}(X)^\vee)$  (or  $\text{Int}(\mathcal{K}(X)^\vee)$  for dual Kähler cone  $\mathcal{K}(X)$ ) contains an element of  $H^{2n-2}(X, \mathbb{Q})$ . Is  $X$  always projective? If not, how algebraic is  $X$  ?

### 3 Moishezon morphisms

Let us first recall the definition of a projective morphism.

**Definition 3.1** (Projective morphism, first definition). Let  $X \rightarrow S$  be a proper morphism between complex spaces.  $f$  is projective if there exists a locally free coherent sheaf  $\mathcal{E}$  of finite rank such that there exists a closed  $S$ -immersion  $X \hookrightarrow \mathbb{P}_S(\mathcal{E})$ , with the following diagram commuting.

$$\begin{array}{ccc} X & \xrightarrow{\quad} & \mathbb{P}_S(\mathcal{E}) \\ & \searrow & \swarrow \\ & S & \end{array}$$

**Definition 3.2** (Projective morphism, second definition). Let  $X \rightarrow S$  be a proper morphism between complex spaces.  $f$  is projective if  $X$  can be embedded in  $\mathbb{P}^N \times S$  for some  $N$ , with the following diagram commuting.

$$\begin{array}{ccc} X & \xrightarrow{\quad} & \mathbb{P}^N \times S \\ & \searrow & \swarrow \\ & S & \end{array}$$

Note that Kollár adopts the second definition.

**Definition 3.3** (Locally projective morphism). Let  $f : X \rightarrow S$  be a proper morphism of complex spaces. We call  $f$  locally projective if for every relatively compact open subset  $Q$  of  $S$  the restriction  $f_Q : X_Q \rightarrow Q$  is a projective morphism.

**Remark 3.4.** It is easy to see the second definition immediately implies the first definition. The converse direction also holds when the base is Stein or quasi-projective.

*Proof.* Assume we have the 1st definition, so that  $f : X \rightarrow S$  and  $g : Y = \mathbb{P}_S(f_* \mathcal{L}^{\otimes m}) \rightarrow S$ . Let  $A$  be an  $g$ -ample line bundle. And, therefore by Serre vanishing theorem over some Stein compact subset  $B \subset S$ , for some sufficiently large  $n \gg 0$ , we have

$$g^* g_*(\mathcal{E} \otimes A^{\otimes n}) \rightarrow \mathcal{E} \otimes A^{\otimes n},$$

is surjective. Since the base  $S$  is Stein, by Cartan A theorem,  $g_*(\mathcal{E} \otimes A^{\otimes n})$  is globally generated. And therefore so is the pull back  $g^* g_*(\mathcal{E} \otimes A^{\otimes n})$ . Since the surjective map sends a globally generated coherent sheaf to a globally generated coherent sheaf, this means that  $\mathcal{E} \otimes A^{\otimes n}$  is globally generated.

By coherence of  $\mathcal{E} \otimes A^{\otimes n}$ , the cohomology group  $V = H^0(Y, \mathcal{E} \otimes A^{\otimes n})$  is finite dimensional. And there is a surjection

$$V \otimes \mathcal{O}_Y \rightarrow \mathcal{E} \otimes A^{\otimes n}.$$

And therefore it will induce an embedding

$$X \hookrightarrow \mathbb{P}_B(\mathcal{E}) = \mathbb{P}_B(\mathcal{E} \otimes A^{\otimes m}) \hookrightarrow \mathbb{P}(V) \times B,$$

after shrinking the base  $B \subset S$ . □

**Remark 3.5.** When the total space has only a finite number of irreducible components, then a locally projective morphism is bimeromorphic to a projective morphism. (see [Fuj83, Lemma 1.3.1]).

In what follows, we may assume that the base  $S$  is reduced. However, in general, we do not require the total space  $X$  to be reduced or not.

**Definition 3.6** (Moishezon morphism, 1st definition). A proper morphism of analytic spaces  $g : X \rightarrow S$  is Moishezon if  $g : X \rightarrow S$  is bimeromorphic to a projective morphism  $g^p : X^p \rightarrow S$ .

That is, there is a closed subspace  $Y \subset X \times_S X^p$  such that the coordinate projections  $Y \rightarrow X$  and  $Y \rightarrow X^p$  are bimeromorphic.

$$\begin{array}{ccc} & Y & \\ \swarrow & & \searrow \\ X & \overset{\text{-----}}{\longrightarrow} & X^p \\ \searrow & & \swarrow \\ & S & \end{array}$$

**Definition 3.7** (Moishezon morphism, 2nd definition). A proper morphism of analytic spaces  $g : X \rightarrow S$  is Moishezon if there is a projective morphism of algebraic varieties  $G : \mathbb{X} \rightarrow \mathbb{S}$  and a meromorphic  $\phi_S : S \dashrightarrow \mathbb{S}$  such that  $X$  is bimeromorphic to  $\mathbb{X} \times_{\mathbb{S}} S$ , the fiber product of rational maps is defined where the maps are defined, so on a dense open set.

**Remark 3.8.** Let us say a few words about the fiber product for a rational map  $\phi_S : S \dashrightarrow \mathbb{S}$ , the fiber product is defined on the place that  $\phi_S$  is a holomorphic map.

**Definition 3.9** (Moishezon morphism, 3rd definition). A proper morphism of analytic spaces  $g : X \rightarrow S$  is Moishezon if there is a rank 1, reflexive sheaf  $L$  on  $X$  such that the natural map  $X \dashrightarrow \text{Proj}_S(g_* L)$  is bimeromorphic onto the closure of its image.



**Proposition 3.10.** The three definitions of Moishezon morphism are equivalent.

*Proof.* Definition 3.7 equivalent to Definition 3.6 is clear (using Proposition 3.16). Conversely, if there exists a projective family  $X^p \rightarrow S$  that is bimeromorphic to a given  $f : X \rightarrow S$ , then by generic flatness we know  $g^p : X^p \rightarrow S$  is flat over  $S^o$  for some Zariski open subset  $S^o \subset S$ , and therefore using the definition of projective family, there exists a morphism

$$S^o \rightarrow \text{Hilb}(\mathbb{P}^N)$$

such that the projective family is the pull back

$$\begin{array}{ccc} X^p & \longrightarrow & \mathcal{M} \\ \downarrow & & \downarrow \\ S & \dashrightarrow & \text{Hilb}(\mathbb{P}^N) \end{array}$$

We now show that the first definition and third definition are equivalent. From the third definition to the first definition is clear since  $\text{Proj}_S(f_*L)$  is projective over  $S$ . Conversely, if  $f : X \rightarrow S$  is bimeromorphic to a projective morphism  $X^p \rightarrow S$ . Then since we assume  $X$  is normal, therefore the meromorphic map  $X \dashrightarrow X^p$  is a morphism outside a codimension 2 subset. And the pull back  $(\phi^o)^*\mathcal{O}_X(1)$  is a big line bundle defined on a big open subset, which can be extended uniquely to a big rank 1 reflexive sheaf.  $\square$

**Remark 3.11.** The terminology in different papers is different; we can summarize it as below.

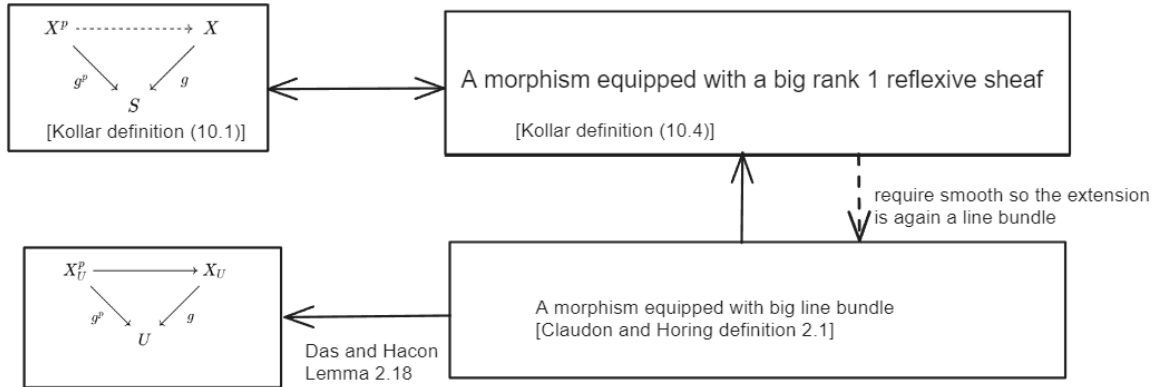


Figure 1: Definitions in different papers

The Moishezon morphism satisfies the following Chow type lemma (which can be viewed as the deterministic property of a Moishezon morphism).

**Theorem 3.12** ([DH20, Lemma 2.18]). Let  $f : X \rightarrow S$  be a proper surjective morphism of analytic varieties, and let  $L$  be a  $f$ -big line bundle on  $X$  and  $D$  a  $\mathbb{Q}$ -divisor. Then

- (1) Over any relatively compact open subset  $V \subset S$ , there exists a proper (indeed it's projective see [CH24]) bimeromorphic morphism  $\alpha : W \rightarrow f^{-1}V$  from a smooth analytic variety  $W$  such that

$\beta = f|_{f^{-1}V} \circ \alpha : W \rightarrow V$  is a projective morphism and,

(2)  $\left(W, \alpha_*^{-1} \left(D|_{f^{-1}V}\right) + \text{Ex}(\alpha)\right)$  is a log smooth pair.

First, let us compare the theorem above with Definition 3.6; in the Definition, we only assume the existence of some bimeromorphic  $S$ -map, whereas the Chow lemma allows us to choose some bimeromorphic projective morphism.

*Proof.* Let  $\phi : X \dashrightarrow Y$  be the relative Iitaka fibration of  $L$  over  $S$  and  $g : Y \rightarrow S$  the induced projective morphism. Since  $L$  is  $f$ -big,  $\phi : X \dashrightarrow Y$  is bimeromorphic. Let  $p : \Gamma \rightarrow X$  and  $q : \Gamma \rightarrow Y$  be the resolution of indeterminacy of  $\phi$  so that  $p$  is proper.

$$\begin{array}{ccc}
 & \Gamma & \\
 p \swarrow & & \searrow q \\
 X & \xrightarrow{\phi} & Y \\
 f \searrow & & \swarrow g \\
 & S &
 \end{array}$$

Now fix a relatively compact open subset  $V \subset S$ . Choose another relatively compact open set  $U \subset S$  containing  $V$  such that  $\bar{V} \subset U$ . Note that  $U$  is  $\sigma$ -compact, since it is relatively compact. Since  $f$  and  $g$  are both proper morphisms, it follows that  $X_U := f^{-1}U$  and  $Y_U := g^{-1}U$  are both  $\sigma$ -compact. Let  $\Gamma_U := q^{-1}(g^{-1}U) = p^{-1}(f^{-1}U)$ . Then from the commutative diagram above it follows that  $q|_{\Gamma_U} : \Gamma_U \rightarrow g^{-1}U$  is a proper morphism. In particular,  $\Gamma_U$  is  $\sigma$ -compact. Note that  $q|_{\Gamma_U}$  is bimeromorphic. Therefore there is a projective bimeromorphic morphism  $h : Z \rightarrow \Gamma_U$  from an analytic variety  $Z$  such that  $q|_{\Gamma_U} \circ h : Z \rightarrow Y_U$  is a projective bimeromorphic morphism. Since  $g$  is projective, so is  $Z \rightarrow U$ .

Now we replace  $U$  by our previously fixed open set  $V$ . Then  $Z_V := (g \circ q \circ h)^{-1}V$  is a relatively compact open subset of  $Z$ . Let  $r : W \rightarrow Z_V$  be the log resolution of  $\left(Z_V, (p \circ h)_*^{-1} \left(D|_{f^{-1}V}\right)\right)$ .

Let  $\alpha := p|_{\Gamma_V} \circ h|_{h^{-1}\Gamma_V} \circ r$  and  $\beta := g|_{g^{-1}V} \circ q|_{\Gamma_V} \circ h|_{h^{-1}\Gamma_V} \circ r$ , where  $\Gamma_V := p^{-1}(f^{-1}V) = q^{-1}(g^{-1}V)$ . Note that  $\beta$  is a projective morphism, since it is a composition of projective morphisms over relatively compact bases.

Then  $\alpha : W \rightarrow f^{-1}V$  is a proper bimeromorphic morphism and  $\beta : W \rightarrow V$  is a projective morphism such that  $\beta = f|_{f^{-1}V} \circ \alpha$  and  $\left(W, \alpha_*^{-1} \left(D|_{f^{-1}V}\right) + \text{Ex}(\alpha)\right)$  is a log smooth pair.  $\square$

**Proposition 3.13** ([Fuj83, Proposition 1.5.(4)]). Suppose that there exists a locally projective morphism  $g : Y \rightarrow S$  and a generically finite meromorphic  $S$ -map  $h : X \dashrightarrow Y$ . Then  $f$  is Moishezon.

$$\begin{array}{ccc}
 X & \xrightarrow{h} & Y \\
 f \searrow & & \swarrow g \\
 & S &
 \end{array}$$

*Proof.* First, since being Moishezon is stable under bimeromorphic change, without loss of generality we can assume that  $h$  is a morphism. Since Moishezon morphism and locally projective morphism are proper,  $h$  is proper. Apply the Stein factorization theorem, such that  $h_2$  is projective (since  $h_2$  is finite) and  $h_1$  is proper. Thus, the composition  $g \circ h_2$  is locally projective. And thus by definition  $X \rightarrow S$  is a Moishezon morphism.

$$\begin{array}{ccc}
 & X^* & \\
 h_1 \nearrow & & \searrow h_2 \\
 X & \xrightarrow{h} & Y \\
 & \searrow & \nearrow g \\
 & S &
 \end{array}$$

□

**Proposition 3.14** ([Fuj83, Proposition 1.7]). Let  $f : X \rightarrow S$  be a Moishezon morphism, and  $g : Y \rightarrow S$  a proper morphism, of reduced complex spaces. Suppose that there is a generically surjective meromorphic  $S$ -map  $h : X \dashrightarrow Y$ . Then  $g$  also is Moishezon.

*Proof.* This Proposition can be viewed as a generalization of Proposition 2.17. The proof is a bit involved, and we omit it here. □

**Proposition 3.15** ([Fuj83, Proposition 1.5]).

(1) The morphism  $f : X \rightarrow S$  is Moishezon if and only if for each irreducible component  $X_i$  of  $X$  the restriction  $f = f|_{X_i} : X_i \rightarrow S$  is Moishezon.

(2) Let  $f : X \rightarrow S$  be a Moishezon morphism. Then: For every reduced analytic subspace  $X' \subseteq X$  the induced morphism  $f' = f|_{X'} : X' \rightarrow S$  is Moishezon.

*Proof.* For (1), let's take the normalization

$$\nu : X^\nu \rightarrow X,$$

recall that for a reduced complex space with finitely many irreducible components, the normalization is a bimeromorphic map. So that  $f : X \rightarrow S$  is Moishezon iff the restrictions on each component  $X_i$  are Moishezon.

For (2), by the Chow lemma (Theorem 3.12), we can find some locally projective morphism such that  $X^*$  is smooth and  $h$  is a bimeromorphic  $S$ -morphism.

$$\begin{array}{ccc}
 X^* & \xrightarrow{h} & X \\
 & \searrow g & \nearrow f \\
 & S &
 \end{array}$$

We then take the inverse image of the analytic subspace  $X'$  and denote it  $Z = h^{-1}(X')$ . (we can assume the inverse image has reduced structure). Since the restriction of the projective morphism on  $g|_Z : Z \rightarrow S$  is still locally projective. And by construction, clearly the morphism  $Z \rightarrow X'$  is surjective. And therefore, by Proposition 3.14, we know that  $X' \rightarrow S$  is a Moishezon morphism. □

Restriction on the image side will also preserve the Moishezon condition.

**Proposition 3.16** (A morphism is Moishezon iff it is Moishezon onto its image). Let  $f : X \rightarrow S$  be a proper morphism between analytic spaces, let  $f' : X \rightarrow f(X) = Y \subset S$  be the restriction, then  $f$  is Moishezon (resp. projective) iff  $f'$  is Moishezon (resp. projective).

*Proof.* It's enough to prove the case for the projective morphism case (and the Moishezon morphism case follows easily).

To see this, assume that the morphism  $f : X \rightarrow S$  is projective, by definition it factors through  $X \hookrightarrow \mathbb{P}_S^n \rightarrow S$ . Doing base change on  $Y \hookrightarrow S$ , proves the projectivity of  $X \hookrightarrow \mathbb{P}_Y \rightarrow Y$ .

The converse direction is clear, since the composition  $X \rightarrow f(X) \hookrightarrow S$  can be written as:

$$X \hookrightarrow \mathbb{P}_{f(X)}^n \rightarrow f(X) \hookrightarrow S.$$

Since  $\mathbb{P}_{f(X)}^n = \mathbb{P}_S^n \times_S f(X)$ , we can rewrite the morphism as:

$$X \hookrightarrow \mathbb{P}_S^n \times_S f(X) \hookrightarrow \mathbb{P}_S^n \rightarrow S,$$

where the second inclusion is because  $f(X) \hookrightarrow S$  and so is the projective bundle.  $\square$

**Proposition 3.17.** When the base is Moishezon, then the total space is Moishezon iff the morphism is Moishezon.

*Proof.* We first prove that a morphism between Moishezon spaces is a Moishezon morphism. Let us define the graph embedding to be

$$\iota : X \rightarrow X \times S, \quad x \mapsto (x, f(x)),$$

since  $X$  is Moishezon it's bimeromorphic to a projective variety, as the diagram below shows

$$\begin{array}{ccccc} X & \xrightarrow{\iota} & X \times S & \dashrightarrow & X^p \times S \\ & \searrow f & \downarrow \pi & \swarrow \pi^p & \\ & & S & & \end{array}$$

Clearly,  $\pi^p$  is a projective morphism. And consequently  $\pi$  is a Moishezon morphism. And finally by Proposition 3.15, the morphism  $f : X \rightarrow S$  is again Moishezon.

Conversely, if the morphism is Moishezon, and  $S$  is a Moishezon space, then there exist bimeromorphic modifications such that the following diagram commutes

$$\begin{array}{ccccc} X^p & \longrightarrow & X' & \longrightarrow & X \\ \downarrow & & \downarrow & \swarrow & \\ S^p & \longrightarrow & S & & \end{array}$$

Where  $X' \rightarrow S$  is a projective morphism and  $S^p$  is a projective variety. Since the base change preserves the projective condition, it is easy to see that  $X^p \rightarrow S^p$  is a projective morphism over  $S^p$ . And therefore  $X^p$  is a projective variety. By Proposition 3.14,  $X'$  is a Moishezon space. Since  $X' \rightarrow X$  is bimeromorphic, this implies that  $X$  is also Moishezon.  $\square$

**Proposition 3.18** ([Kol22, Lemma 15]). Let  $g : X \rightarrow S$  be a proper, generically finite, dominant morphism of normal, complex, analytic spaces. Then  $\text{Ex}(g) \rightarrow S$  is Moishezon.

*Proof.* We will prove the result under the additional assumption that  $S$  is Stein. By the geometric Noether normalization theorem, there exists a finite morphism

$$S \rightarrow \mathbb{C}^{\dim S}.$$

After replacing the base by  $\mathbb{C}^{\dim S}$ , we can assume that the smooth locus of  $S$  is dense in  $g(\text{Ex}(g))$ . Note that, by Proposition 3.13, if the restriction on  $\mathbb{C}^{\dim S}$  is Moishezon, then so will the restriction on  $S$ . We will prove the result by induction on dimension.

We first define the base case  $(g_0 : X_0 \rightarrow S_0) := (g : X \rightarrow S)$ . Let  $E_0$  be a  $g_0$  exceptional divisor, with the image  $Z_0 = g_0(E_0)$ . We then inductively define the morphism  $g_{i+1} : X_{i+1} \rightarrow S_{i+1}$  as follows. Assume that we already constructed  $g_i : X_i \rightarrow S_i$ ; we then blow up  $S_i$  along  $Z_i$  and let  $S_{i+1}$  be the normalization of the blow-up  $\text{Bl}_{Z_i} S_i$ . Since  $S_i$  is reduced, this will induce a generically finite map  $\phi : X_i \dashrightarrow S_{i+1}$ . So that by the universal property of the normalization, the generically finite morphism  $g_i : X_i \rightarrow S_i$  lifts to a generically finite morphism  $g_{i+1} : X_{i+1} \rightarrow S_{i+1}$ , where  $X_{i+1}$  is the normalization of the graph of the map  $\phi : X_i \dashrightarrow S_{i+1}$ .

$$\begin{array}{ccc} X_{i+1} & \xrightarrow{g_{i+1}} & S_{i+1} = \text{Bl}_{Z_i}(S_i)^\nu \\ \downarrow & \nearrow & \downarrow \\ \Gamma_\phi & \xrightarrow{\phi} & \text{Bl}_{Z_i}(S_i) \\ \downarrow & \nearrow & \downarrow \\ X_i & \xrightarrow{g_i} & S_i \end{array}$$

Let  $E_{i+1} \subset X_{i+1}$  denote the bimeromorphic transform of  $E_i$ . (Note that  $X_{i+1} \rightarrow X_i$  is an isomorphism over an open subset of  $E_i$ ). We then compute the vanishing order  $a(E_i, S_i)$  of the Jacobian of  $g_i$  along  $E_i$ . We claim that

$$a(E_{i+1}, S_{i+1}) \leq a(E_i, S_i) + 1 - \text{codim}(Z_i \subset S_i).$$

Thus eventually we reach the situation when  $\text{codim}(Z_i \subset S_i) = 1$ , indeed if  $\text{codim}(Z_i \subset S_i) \geq 2$  then the Jacobian of  $g_i$  along  $E_i$  will eventually go to zero. Contradiction.

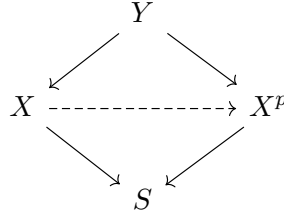
Thus by comparing the dimension we know when we restrict the morphism  $X_i \rightarrow S_i$  to  $E_i \rightarrow Z_i$  it will become a generically finite morphism. Since  $S_{i+1} \rightarrow S_i$  is projective, the composition  $Z_i \rightarrow Z_0$  will be a locally projective morphism.

$$\begin{array}{ccc} E_i & \xrightarrow{\quad} & Z_i \\ & \searrow & \swarrow \\ & Z_0 & \end{array}$$

By Proposition 3.13, we know that  $E_i \rightarrow Z_0$  is a Moishezon morphism. Since the strict transform  $E_i \rightarrow E_0$  is a dominant morphism, by Proposition 3.14, we know that  $E_0 \rightarrow Z_0$  is also a Moishezon morphism. Finally, by Proposition 3.15 and Proposition 3.16, we know that  $\text{Ex}(f) \rightarrow S$  is Moishezon.  $\square$

**Theorem 3.19** (Fibers of the Moishezon morphism are Moishezon spaces, [Kol22, Corollary 16]). The fibers of a proper, Moishezon morphism are Moishezon.

*Proof.* Let  $g : X \rightarrow S$  be a proper, Moishezon morphism. It is bimeromorphic to a projective morphism  $X^p \rightarrow S$ . We may assume  $X^p$  to be normal. Let  $Y$  be the normalization of the closure of the graph of  $X \dashrightarrow X^p$ .



Fix now  $s \in S$ . Let  $Z_s \subset X_s$  be an irreducible component; since given a proper dominant morphism, there exists at least one irreducible component dominating the base, there exists  $W_s \subset Y_s$  an irreducible component that dominates  $Z_s$ . And by Proposition 3.14 and Proposition 3.15, it's enough to show that  $W_s$  is Moishezon. We divide the problem into two cases:

If  $\pi : Y \rightarrow X^p$  is generically an isomorphism along  $W_s$ , then  $W_s$  is bimeromorphic to an irreducible component of  $X_s^p$ , hence Moishezon.

Otherwise  $W_s \subset \text{Ex}(\pi)$ . Now  $\text{Ex}(\pi) \rightarrow X^p$  is Moishezon by Proposition 3.18. And by induction on dimension, since  $\dim \text{Ex}(\pi) < \dim X = \dim Y$ , the fiber  $W_s$  is Moishezon.  $\square$

**Proposition 3.20** ([Kol22, Example 13]). Let  $Z$  be a normal, projective variety with discrete automorphism group. Let  $g : X \rightarrow S$  be a fiber bundle with fiber  $Z$  over a connected base  $S$ . Then  $g$  is Moishezon  $\Leftrightarrow g$  is projective  $\Leftrightarrow$  the monodromy is finite.

**Remark 3.21.** The monodromy here is different from the cohomological monodromy. Here the monodromy is referred to as the fiber bundle monodromy

$$\rho : \pi_1(S) \rightarrow G \subset \text{Aut}(Z)$$

where  $G \subset \text{Aut}(Z)$  is the structure group of the fiber (e.g. when the fiber bundle is a principal  $G$ -bundle, then the structure group is simply the group  $G$ ). Finite monodromy condition means that  $\text{im}(\rho) \subset G$  is a finite subgroup.

Before proving the theorem, let us state a lemma from fiber bundle theory that is useful in what follows (which can be viewed as a generalization of the Ehresmann theorem over a simply connected base: when the base is simply connected, the fiber bundle is automatically trivial).

**Lemma 3.22.** Let  $g : X \rightarrow S$  be a fiber bundle with trivial monodromy group, then the fiber bundle is actually a trivial fiber bundle.

The following proposition about the automorphism of a polarized projective variety will be useful in the proof.

**Proposition 3.23.** Let  $Z$  be a projective variety, and  $L$  an ample line bundle over  $Z$ . Then  $\text{Aut}(Z, L) = \{\phi \in \text{Aut}(Z) \mid c_1(\phi^*L) = c_1(L)\}$  is finite if the automorphism group  $\text{Aut}(Z)$  is discrete.

*Proof.* Let

$$\phi : Z \rightarrow Z \in \text{Aut}(Z, L),$$

consider the graph  $\Gamma_\phi \subset Z \times Z$ , thus the Hilbert polynomial of  $\Gamma_\phi$  relative to  $L \boxtimes L$  is given by

$$H_\phi(n) = \chi(\Gamma_\phi, (L \boxtimes L)^n) = \chi(Z, L^{\otimes n} \otimes \phi^*(L^{\otimes n})).$$

On the other hand, since  $c_1(\phi^*L) = c_1(L)$ , it is easy to see that  $L^{\otimes n} \otimes \phi^*(L^{\otimes -n})$  is numerically trivial, and thus

$$H(n) = \chi(Z, L^{\otimes n} \otimes \phi^*(L^{\otimes n}) \otimes L^{\otimes n} \otimes \phi^*(L^{\otimes -n})) = \chi(Z, L^{\otimes 2n}),$$

which is independent of  $\phi$  and denote it  $P(n) = H_\phi(n)$ . So that the graph lies on  $\text{Hilb}_{Z \times Z}^P$  (with fixed Hilbert polynomial  $P(n)$ ), which is of finite type. Thus it contains finitely many irreducible components (by the Noetherian property). On the other hand, since  $Z$  is projective, each irreducible component of the Hilbert scheme is proper. Thus  $\text{Aut}(Z, L)$  is finite.  $\square$

The idea of the proof of the theorem is provided by Professor Kollár.

*Proof of the theorem.* We only need to show that (1) implies (3) and (3) implies (2). For (3) implies (2), we try to take an étale base change so that the fiber bundle becomes a trivial bundle. We can do this as follows. Consider  $\rho(\pi_1(S)) \subset \text{Aut}(Z)$  is finite, let  $S' \rightarrow S$  be the corresponding finite (unbranched) cover that kills the monodromy. Indeed since we have

$$\rho : \pi_1(S) \rightarrow G$$

then the kernel  $\ker(\rho)$  is a subgroup of  $\pi_1(S)$  of finite index, therefore by the Galois correspondence for coverings, there exists a finite étale cover of the base

$$\tilde{S} \rightarrow S,$$

such that the monodromy of the fiber bundle under the base change becomes trivial, then by the previous lemma, after the base change the fiber bundle becomes a trivial bundle

$$Z \times \tilde{S} \rightarrow \tilde{S},$$

clearly the morphism is projective and admits a relative ample line bundle  $L$  (since  $Z$  is projective). And therefore if we define

$$L' = \bigotimes_{g \in \Gamma} g^*L.$$

Since  $L'$  is monodromy invariant, the ample line bundle will descend to the original fiber bundle  $g : X \rightarrow S$  and thus  $g$  is a projective morphism.

For (1) implies (3), since  $g : X \rightarrow S$  is Moishezon, by Definition 3.9, there exists a  $g$ -big (rank 1 reflexive sheaf)  $H$  on  $X$  (since it's a fiber bundle, the restriction on  $Z$  is again big, and we denote it also as  $H$ ). Given an ample line bundle  $L$  on the fiber  $Z$ , we consider the monodromy action on  $L$ , which pulls back the ample line bundle to another ample line bundle  $L_\gamma = \rho(\gamma)^*L$ .

Note that under the monodromy action, the intersection

$$d := H \cdot (L_\gamma)^{n-1},$$

remains the same for all  $\gamma$ .

We then consider the linear functional

$$\ell : N^1(Z)_\mathbb{R} \rightarrow \mathbb{R}, \quad M \mapsto M^{n-1} \cdot H,$$

if we restrict the linear functional on the ample cone  $\text{Amp}(Z)$ , then

$$S_d = \{M \in N^1(Z)_\mathbb{R} \cap \text{Amp}(Z) \mid M^{n-1} \cdot H = d\},$$

is a bounded slice. To see this, by the Khovanskii-Teissier inequality we have

$$(H \cdot M^{n-1})^n \geq (H^n) (M^n)^{n-1},$$

thus we get

$$\text{vol}(M) \leq \left( \frac{d^n}{H^n} \right)^{\frac{1}{n-1}},$$

so that it is bounded (for a fixed  $H$ ). Thus the slice contains only finitely many lattice points of  $\text{NS}(Z)$ .

$$\#\{M \in \text{NS}(Z) \cap \text{Amp}(Z) \mid \ell(M) = d\} < \infty.$$

In particular, the ample line bundle on the monodromy orbit is finite

$$\Gamma \cdot L = \{L_\gamma := \rho(\gamma) \cdot L \mid \gamma \in \pi_1(S)\} \subset \{M \in \text{NS}(Z) \cap \text{Amp}(Z) \mid \ell(M) = d\}.$$

This will force the monodromy to be finite, indeed apply the orbit-stabilizer theorem

$$\boxed{|\Gamma| = |\Gamma \cdot L| |\text{Stab}(L)| < +\infty}$$

Thus we only need to prove that  $\text{Stab}(L) = \text{Aut}(Z, L) = \{\phi \in \text{Aut}(Z) \mid \phi^*L = L\}$  is finite. On the other hand, since  $\text{Aut}(Z)$  is discrete, by Proposition 3.23, this means that  $\text{Stab}(L) = \text{Aut}(Z, L)$  is finite.  $\square$

## 4 Examples

In this section, we will present various examples related to the Moishezon space and Moishezon morphism.



## 4.1 The Hironaka's example

Hironaka discovered a bunch of complete non-projective 3-folds, which are called Hironaka's varieties. Note that based on the construction of Hironaka, we can from almost all the projective varieties construct some Moishezon spaces, which is why we said at the beginning that Moishezon spaces are versatile in birational geometry. (However, this is not true in dimension 2, since all the smooth Moishezon surfaces are actually projective, see e.g. [GPR94]). The major reference of this part of the note is the paper by Ulrich Thiel (see [https://ulthiel.com/math/wp-content/uploads/other/hironakas\\_example.pdf](https://ulthiel.com/math/wp-content/uploads/other/hironakas_example.pdf)).

Given a smooth projective threefold, which contains two rational curves transversely intersecting at two points. Assume that the two rational curves are  $C$  and  $D$  that intersect at the points  $P, Q$ .

We then take two steps, blow up

$$\begin{aligned} X_1 &= \text{Bl}_{(D \setminus P)'} (\text{Bl}_{C \setminus P}(X \setminus P)) \xrightarrow{\pi_2} \text{Bl}_{C \setminus P}(X \setminus P) \xrightarrow{\pi_1} X \setminus P \\ X_2 &= \text{Bl}_{(C \setminus Q)'} (\text{Bl}_{D \setminus Q}(X \setminus Q)) \xrightarrow{\sigma_2} \text{Bl}_{D \setminus Q}(X \setminus Q) \xrightarrow{\sigma_1} X \setminus Q, \end{aligned}$$

Note that if we define  $U = X - \{P, Q\}$ , then  $\pi^{-1}(U) \cong \sigma^{-1}(U)$ . In particular, we can glue  $X_1$  and  $X_2$  along  $\pi^{-1}(U)$  and  $\sigma^{-1}(U)$ . In the picture below, we glue the red exceptional surface on the right hand side with the black exceptional surface on the left hand side (denote it  $S_1$ ) and the blue exceptional surface on the left hand side with the black exceptional surface on the right hand side (denote it  $S_2$ ). (see picture 2). By the gluing lemma, there exists a morphism  $f : H \rightarrow X$  and the restriction of the morphism on  $S_1, S_2$  as  $f_1 = f|_{S_1} : S_1 \rightarrow C$  and  $f_2 = f|_{S_2} : S_2 \rightarrow C$ .

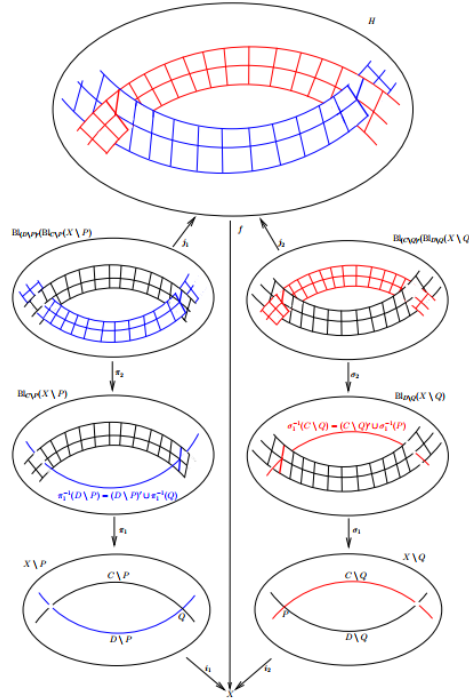


Figure 2: Construction of the Hironaka's variety

We claim that Hironaka's variety is non-projective. The idea to prove the non-projectivity is to find some curve on the surface  $S = S_1 \cup S_2$  which has positive degree but adds up to 0.

The key observation is that  $f^{-1}(P)$  (resp.  $f^{-1}(Q)$ ) decomposes into two split projective lines  $L_Q$  and  $L'_Q$  in  $S_1$  (resp.  $L_P$  and  $L'_P$  in  $S_2$ ). (see the precise statement below).

Choose two points  $A \in C - \{P, Q\}$  and  $B \in D - \{P, Q\}$ . Since all the points on a rational curve are linearly equivalent, therefore

$$\begin{aligned} A \sim_C Q &\implies f_1^{-1}(A) \sim_{S_1} f_1^{-1}(Q) = L_Q + L'_Q \\ B \sim_D P &\implies f_2^{-1}(B) \sim_{S_2} f_2^{-1}(P) = L_P + L'_P \end{aligned}$$

and pushing forward the cycle, we get equivalence on  $S$ .

$$\begin{aligned} I : f^{-1}(A) &\sim_S f^{-1}(Q) = L_Q + L'_Q \\ II : f^{-1}(B) &\sim_S f^{-1}(P) = L_P + L'_P \end{aligned}$$

On the other hand we also have that  $B, Q$  lie on the same rational curve, so that

$$III : B \sim_D Q \implies f_2^{-1}(B) \sim_{S_2} f_2^{-1}(Q) \implies f^{-1}(B) \sim_S L'_Q$$

and combining them together, we get

$$\begin{aligned} f^{-1}(A) + f^{-1}(B) &\sim_S f^{-1}(A) + f^{-1}(B) \implies L_Q + L'_Q + L_P + L'_P \sim_S L'_Q + L'_P \\ &\implies L_Q + L_P \sim_S 0 \end{aligned}$$

If there exists some ample divisor on  $A$ , then both  $L_Q \cdot A > 0$  and  $L_P \cdot A > 0$  contradict the linearly trivial relation above. Therefore the only possible case is that Hironaka's variety is non-projective.

## 4.2 Flop the lines on general quintic threefold produce Moishezon variety

### 4.3 Locally Moishezon morphism which is not globally Moishezon

There are rational and K3 surfaces with infinite, discrete automorphism group. These lead to fiber bundles over the punctured disc  $\mathbb{D}^\circ$  that are locally Moishezon but not globally Moishezon (using Proposition 3.22).

### 4.4 Singular Kähler Moishezon space need not be projective

By blowing down elliptic curves, such an easy example is not possible. Instead, consider a cubic  $C \subset \mathbb{P}_2$  and let  $x_1, \dots, x_{10}$  be general points on  $C$ . Let  $f : X \rightarrow \mathbb{P}_2$  be the blow-up of these points. Then the strict transform  $\hat{C}$  of  $C$  in  $X$  is elliptic with  $\hat{C}^2 = -1$ . It can be shown that the blow-down of  $\hat{C}$  is not projective.

### 4.5 Fiberwise projective morphism need not be a projective morphism

Let  $S_0 := (g = 0) \subset \mathbb{P}_x^3$  and  $S_1 := (f = 0) \subset \mathbb{P}_x^3$  be surfaces of the same degree. Assume that  $S_0$  has only ordinary nodes,  $S_1$  is smooth Pic( $S_1$ ) is generated by the restriction of  $\mathcal{O}_{\mathbb{P}^3}(1)$  and  $S_1$

does not contain any of the singular points of  $S_0$ . Fix  $m \geq 2$  and consider

$$X_m := (g - t^m f = 0) \subset \mathbb{P}_x^1 \times \mathbb{A}_t^1.$$

The singularities are locally analytically of the form  $xy + z^2 - t^m = 0$ . Thus  $X_m$  is locally analytically factorial if  $m$  is odd. If  $m$  is even then  $X_m$  is factorial since the general fiber has Picard number 1, but it is not locally analytically factorial; blowing up  $(x = z - t^{m/2} = 0)$  gives a small resolution. Thus we get that (4.1)  $X_m$  is bimeromorphic to a proper, smooth family of projective surfaces iff  $m$  is even, but (4.2)  $X_m$  is not bimeromorphic to a smooth, projective family of surfaces.

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## 1 Overview

The aim of this note is to study fiberwise bimeromorphic problems. To be more concrete, we consider the following two problems. The first one is:

**Question 1.1.** Let

$$\pi : X \rightarrow B \text{ and } \pi' : X' \rightarrow B,$$

be proper flat morphisms from a complex analytic space to a smooth connected curve  $B$ . Assume that the generic fibers of  $\pi$  and  $\pi'$  are bimeromorphic. Under what conditions do the special fibers between these two families also admit a certain bimeromorphic relation?

The second one focuses on the Moishezon morphisms, under which condition we can let a Moishezon morphism be fiberwise bimeromorphic to a projective morphism:

**Question 1.2.** Let  $g : X \rightarrow \mathbb{D}$  be a flat, proper, Moishezon morphism. Under which conditions is it actually fiberwise bimeromorphic to a projective morphism  $g^p : X^p \rightarrow \mathbb{D}$ ?

We will discuss the first question in Sections 2–4 and the second question in Section 5. The main references for this note are [Kol22] and [KT19] and [CRT25].

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## 2 A Fiberwise Birational Criterion

We first recall the definition of the meromorphic  $S$  map that we introduced for the first time.

**Definition 2.1** (Meromorphic  $S$ -map). Let  $X, Y$  be reduced complex spaces. We call the  $S$ -map a *meromorphic  $S$ -map*

$$\begin{array}{ccc}
 & \Gamma & \\
 p \swarrow & & \searrow q \\
 X & \overset{\alpha}{\dashrightarrow} & Y \\
 f \searrow & & \swarrow g \\
 & S &
 \end{array}$$

if there exists a subvariety  $\Gamma \subset X \times_S Y$  with the restriction of the first projection  $p : \Gamma \rightarrow X$  being a proper bimeromorphic morphism. Moreover, if the restriction on the second projection  $q : \Gamma \rightarrow Y$  is also a proper bimeromorphic morphism, then we call  $\alpha$  *proper bimeromorphic  $S$ -map*.

**Definition 2.2** (Fiberwise bimeromorphic map, [Kol22, Definition 26]). Let  $g_i : X^i \rightarrow S$  be proper morphisms. A bimeromorphic  $S$ -map  $\phi : X^1 \dashrightarrow X^2/S$  is *fiberwise bimeromorphic* if  $\phi$  induces a bimeromorphic map  $\phi_s : X_s^1 \dashrightarrow X_s^2$  for every  $s \in S$ .

**Remark 2.3** (Fiberwise bimeromorphic  $\neq$  fibers bimeromorphic equivalent, [CRT25, Example 2.15]). Let  $\mathbb{F}_n = \mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(n)) \rightarrow \mathbb{P}^1$  be the Hirzebruch surface of index  $n$ . By construction, it is easy to see that all the Hirzebruch surfaces are birationally equivalent to  $\mathbb{P}^1 \times \mathbb{P}^1$ . Let  $Z$  be any compact complex manifold. So that  $\mathbb{F}_n \times Z \rightarrow \mathbb{P}^1$  is birationally equivalent to  $(\mathbb{P}^1 \times \mathbb{P}^1) \times Z \rightarrow \mathbb{P}^1$ .

$$\begin{array}{ccccc}
 & \text{Bl}_{p_1, \dots, p_n}(\mathbb{P}^1 \times \mathbb{P}^1) \times Z & & & \\
 p \swarrow & & & & \searrow q \\
 \mathbb{P}^1 \times \mathbb{P}^1 \times Z & \overset{q \circ p^{-1}}{\dashrightarrow} & \mathbb{F}_n \times Z & & \\
 & \searrow & \swarrow & & \\
 & \mathbb{P}^1 & & &
 \end{array}$$

Note that fibers of these two families are birationally equivalent (as both sides have fiber  $\mathbb{P}^1 \times Z$ ). However the restriction of the map  $q \circ p^{-1}$  does not give the bimeromorphic map of the fiber (since the strict transform of the fibers of  $\mathbb{P}^1 \times \mathbb{P}^1 \times Z$  via  $p^{-1}$  will be contracted by  $q$ ).

Although the bimeromorphic map need not be fiberwise bimeromorphic in general, it is indeed fiberwise bimeromorphic on a dense open subset, as the following proposition shows.

**Proposition 2.4** (Bimeromorphic on the generic fiber). Let  $f : X \dashrightarrow Y$  be a bimeromorphic  $S$ -map between two proper surjective families  $g : X \rightarrow S$ ,  $h : Y \rightarrow S$  over the base  $S$ . Then on the generic fiber, the map  $f$  will induce a bimeromorphic  $f_s : X_s \dashrightarrow Y_s$ .



Note that by definition, both of the  $p_i$  are open embeddings on  $Y \setminus (\text{Ex}(p_1) \cup \text{Ex}(p_2))$ .

Thus if we prove that neither  $p_1(\text{Ex}(p_1) \cup \text{Ex}(p_2))$  nor  $p_2(\text{Ex}(p_1) \cup \text{Ex}(p_2))$  contains a fiber of  $f_1$  or  $f_2$ , then  $p_2 \circ p_1^{-1} : X^1 \dashrightarrow X^2$  (which need not be birational) restricts to a birational map  $X_b^1 \dashrightarrow X_b^2$  for every  $b \in B$ .

We may assume that  $B$  is affine (as we only care about the special fiber, thus we can focus on the affine base around  $b$ ) and let  $\text{Bs } |mK_{X^i}|$  denote the set-theoretic base locus. Let  $L_i = \mathcal{O}_{X^i}(mK_{X^i})$ . The direct image  $\mathcal{E}_i = f_{i*}L_i$ , being a torsion-free sheaf, is locally free on the smooth curve  $B$  (so that the vanishing locus of a section of  $\mathcal{E}_i$  is a subvariety).

By assumption  $|mK_{X_b^i}| \neq \emptyset$  for generic  $b \in B$ , we claim that  $|mK_{X^i}|$  is non-empty as well. Indeed, the restriction map

$$H^0(X^i, mK_{X^i}) \rightarrow H^0(X_b^i, mK_{X_b^i}),$$

is surjective on the generic fibers. Thus, the pluricanonical system on the generic fiber satisfies  $|mK_{X^i}|_{X_b^i} = |mK_{X_b^i}| \neq \emptyset$ . In particular, this means that  $|mK_{X^i}| \neq \emptyset$ .

On the other hand, since  $H^0(X, mK_{X^i}) \neq 0$  and the base is affine, for any point  $s \in B$ , there exists a non-zero section

$$0 \neq \sigma \in H^0(B, \mathcal{E}_i),$$

such that  $\sigma(s) \neq 0$ . Therefore consider the restriction commutative diagram (note that in general it is not clear whether the base change morphism  $H^0(X_s^i, L_i|_{X_s^i}) \rightarrow \mathcal{E}_i(s)$  is an isomorphism or not on the special fiber)

$$\begin{array}{ccc} H^0(X^i, L_i) & \xrightarrow{\text{res}} & H^0(X_s^i, L_i|_{X_s^i}) \\ \simeq \downarrow & & \downarrow \\ H^0(B, \mathcal{E}_i) & \xrightarrow{\text{res}} & \mathcal{E}_i(s) \end{array}$$

there exists a section  $\sigma' \in H^0(X^i, L_i)$  which maps down to  $\sigma \in H^0(B, \mathcal{E}_i)$  such that  $\sigma(s) \neq 0$ . So that  $\sigma'|_{X_s^i} \neq 0$  in  $H^0(X_s^i, L_i|_{X_s^i})$ . And therefore the base locus  $\text{Bs } |mK_{X^i}|$  cannot contain the fiber.

Since  $X^i$  are smooth,

$$K_Y \sim p_i^* K_{X^i} + E_i, \quad \text{where } E_i \geq 0 \text{ and } \text{Supp } E_i = \text{Ex}(p_i).$$

So that every section of  $\mathcal{O}_Y(mK_Y)$  pulls back from  $X^i$ . Thus

$$\text{Bs } |mK_Y| = p_i^{-1}(\text{Bs } |mK_{X^i}|) + \text{Supp } E_i,$$

Comparing these for  $i = 1, 2$ , we conclude that

$$p_1^{-1}(\text{Bs } |mK_{X^1}|) + \text{Supp } E_1 = p_2^{-1}(\text{Bs } |mK_{X^2}|) + \text{Supp } E_2,$$

Therefore,

$$\boxed{p_1(\text{Supp } E_2) \subset p_1(\text{Supp } E_1) + \text{Bs } |mK_{X^1}|}$$

Since  $E_1$  is  $p_1$ -exceptional,  $p_1(\text{supp } E_1)$  has codimension  $\geq 2$  in  $X^1$ , hence it does not contain any of the fibers of  $f_1$ . Combined with the fact that  $\text{Bs } |mK_{X^1}|$  does not contain any of the fibers either.

Thus,  $p_1(\text{Ex}(p_1) \cup \text{Ex}(p_2))$  does not contain any of the fibers, and the same argument shows for  $p_2(\text{Ex}(p_1) \cup \text{Ex}(p_2))$ .  $\square$

As remarked by [Kol23], the result holds true even when the pluricanonical systems are empty. That is what we will prove in the next section.

### 3 Kontsevich-Tschinkel's Fiberwise Birational Theorem

**Theorem 3.1** ([KT19, Theorem 1]). Let

$$\pi : X \rightarrow B \text{ and } \pi' : X' \rightarrow B$$

be smooth proper morphisms to a smooth connected curve  $B$ , over a field of characteristic zero. Assume that the generic fibers of  $\pi$  and  $\pi'$  are birational over the function field of  $B$ . Then, for every closed point  $b \in B$ , the fibers of  $\pi$  and  $\pi'$  over  $b$  are birational over the residue field at  $b$ .

We first introduce some new notions that are needed in the proof.

**Definition 3.2** (semi-ring). A *semi-ring*  $(S, +, \times)$  consists of a set  $S$  equipped with two binary operations  $+, \times$ , such that  $+$  makes  $S$  a commutative monoid (which need not be an Abelian group compared to the definition of a ring).

**Definition 3.3** (Burnside semi-ring over a field  $k$ , [KT19, Definition 2]). The *Burnside semi-ring*  $\text{Burn}_+(k)$  of a field  $k$  is the set of  $\sim_k$  equivalence classes of smooth schemes of finite type over  $k$  endowed with a semi-ring structure where multiplication and addition are given by disjoint union and product over  $k$ . (Here the  $\sim_k$  equivalence of two schemes  $X, X'$  is defined as follows:  $X/k \sim_k X'/k$  if and only if  $X$  and  $X'$  are  $k$ -birational.) To be more precise, the addition and multiplication of the semi-ring structure are defined as follows:

- (a) Addition: Disjoint union  $[X] + [Y] = [X \sqcup Y]$ .
- (b) Multiplication: Cartesian product  $[X] \cdot [Y] = [X \times Y]$ .

We then introduce the Grothendieck ring, and we denote by  $\text{Burn}(k)$  the Grothendieck ring generated by  $\text{Burn}_+(k)$ .

**Definition 3.4** (The Grothendieck ring  $\text{Burn}(k)$ ). The *Grothendieck ring*  $\text{Burn}(k)$  that is associated to the Burnside semi-ring  $\text{Burn}_+(k)^+$  is defined as the set of equivalence classes of pairs  $([X], [Y])$ , where  $[X], [Y] \in \text{Burn}_+(k)^+$ . Intuitively,  $([X], [Y])$  represents the "difference"  $[X] - [Y]$ . With the equivalence relation: We say  $([X], [Y]) \sim ([X'], [Y'])$  if there exists  $[Z] \in \text{Burn}_+(k)^+$  such that:

$$[X] + [Y'] + [Z] = [X'] + [Y] + [Z].$$

The ring operations are defined as follows

- (a) Addition:  $([X], [Y]) + ([X'], [Y']) = ([X] + [X'], [Y] + [Y'])$ .
- (b) Multiplication:  $([X], [Y]) \cdot ([X'], [Y']) = ([X \times X'] + [Y \times Y'], [X \times Y'] + [Y \times X'])$ .

**Remark 3.5.** The reason to introduce the Grothendieck ring over the Burnside semi-ring is that it allows one to implement formal subtraction, cut and paste operations.



**Remark 3.6.** Note that we can decompose

$$\text{Burn}(k) = \sqcup_{n \geq 0} \text{Bir}_n(k),$$

where  $\text{Bir}_n(k)$  denotes the  $k$ -birational equivalence class of smooth varieties of dimension  $n$ . Each class can be denoted by  $[L/k]$  with  $L = k(X)$ .

**Proposition 3.7** (Existence of SNC model). Let  $R$  be a complete dvr, let  $K$  be the fractional field (i.e. the generic point of  $\text{Spec}(R)$ ) and  $k$  the residue field (i.e. the special point of  $\text{Spec}(R)$ ). Let  $X/K$  be a geometrically connected smooth proper variety defined over  $K$ , then there exists a regular flat separated  $R$ -scheme of finite type  $\mathcal{X}$ , endowed with an isomorphism of  $K$ -scheme  $\mathcal{X}_K \rightarrow X$  such that the special fiber  $\mathcal{X}_k$  is a divisor with strict normal crossing. We call  $\mathcal{X}/R$  an SNC model of  $X/K$ .

*Proof.* Let us first briefly sketch out the idea. We first reduce the problem to the projective case.  $\square$

**Remark 3.8.** The SNC model also plays an important role in the

**Definition 3.9** (Specialization map, [KT19, (3.2)]). Let  $\mathfrak{o}$  be a complete dvr, let  $K$  be the fractional field (i.e. the generic point of  $\text{Spec}(\mathfrak{o})$ ) and  $k$  the residue field (i.e. the special point of  $\text{Spec}(\mathfrak{o})$ ). We define

$$\rho_n : \text{Bir}_n(K) \rightarrow \mathbb{Z}[\text{Bir}_n(k)],$$

as follows: given a smooth projective family  $X_K \rightarrow \text{Spec}(K)$  (with the function field  $L := K(X_K)$ ), choose one of the family

$$\pi : X \rightarrow \text{Spec}(\mathfrak{o}),$$

where  $\pi$  is proper, such that the generic fiber is  $X_K$  and the special fiber

$$X_0 = \bigcup_{i \in I} d_i D_i,$$

is a SNC divisor, with the strata  $D_J := \bigcap_{j \in J} D_j$ . We then define the specialization map to be

$$\boxed{\rho_n([L/K]) := \sum_{\emptyset \neq J \subseteq I} (-1)^{\#J-1} [D_J \times \mathbb{A}^{\#J-1}/k]}$$

One of the main difficulties in the proof is verifying that the specialization map  $\rho_n : \text{Bir}_n(K) \rightarrow \mathbb{Z}[\text{Bir}_n(k)]$  is well-defined (i.e., it does not depend on the choice of the family  $X \rightarrow D$ ) or representative  $X$  in  $\text{Bir}_n(X)$ . We omit the proof of this part; for details, see [KT19, Theorem 4].

*Proof of Theorem 3.1.* We first reduce the problem to  $\text{Spec}$  of a complete dvr. Let  $\pi : X \rightarrow B$  be a smooth proper morphism to a smooth connected curve  $B$  over  $k$  with fiber  $X$  over the generic point of  $B$ . Let  $K = k(B)$  be the function field of  $B$ . Let  $\kappa_b$  be the residue field at  $b$ , a finite extension of  $k$ . Let  $K_b$  be the completion of  $K$  at  $b$ . Then  $K_b$  is a local field with residue field  $\kappa_b$ . Let

$$\phi_b : K \rightarrow K_b,$$

be the canonical inclusion. By functoriality, it defines a homomorphism

$$\phi_{b,*} : \text{Burn}(K) \rightarrow \text{Burn}(K_b).$$

We then consider the specialization map over the complete dvr  $K_b$ . Note that we have the specialization homomorphism

$$\rho : \text{Burn}(K_b) \rightarrow \text{Burn}(\kappa_b),$$

and the following identity

$$[X_b/\kappa_b] = \rho(\phi_{b,*}([X/K])),$$

which follows immediately from Definition 3.9 of  $\rho$ , since the special fiber is smooth and irreducible. This shows that the birational type of the special fiber is determined by the birational type of the generic fiber.  $\square$

## 4 Fiberwise Bimeromorphic Criterion using Plurigenera

In this section, we will give a criterion for the fiberwise bimeromorphic map using plurigenera. For readers who want to know more about this, please refer to [CRT25].

**Lemma 4.1** ([GPR94, Theorem 1.19]). Let  $f : X \rightarrow Y$  be a proper surjective holomorphic map, assume that  $X$  is reduced and irreducible. Then the set

$$\{y \in Y \mid \dim_x X_y > \dim X - \dim Y \text{ for some } x \in X_y\}.$$

is analytic in  $Y$  and of codimension at least 2.

**Proposition 4.2** ([CRT25, Theorem 1.4]). Let  $X, Y$  and  $S$  be complex analytic spaces. Assume that  $X$  is reduced (not necessarily normal) and irreducible,  $Y$  is normal, and  $S$  is a smooth curve. Assume further that both  $\pi_1 : X \rightarrow S$  and  $\pi_2 : Y \rightarrow S$  are proper surjective holomorphic maps. Suppose that there is a bimeromorphic morphism  $f : X \rightarrow Y$  over  $S$ . For some  $t \in S$ , if  $D_t$  is an irreducible component of  $Y_t$  that is of codimension 1 in  $Y$ , then there exists an irreducible component  $C_t$  (equipped with the reduced structure) of  $X_t$  that is bimeromorphic to  $D_t$ , induced by  $f$ .

In particular, if the fibers of  $X \rightarrow S$  and  $Y \rightarrow S$  are irreducible then  $f$  is a fiberwise bimeromorphic map.

*Proof.* Since  $X$  is reduced and irreducible, by the lemma above, we have that the set of points where  $\dim X_y = 0$  is a big open subset of  $Y$  (with the complement an analytic subset  $V$  such that  $\text{codim}_Y(V) \geq 2$ ). Since  $Y$  is normal, and  $f : X - f^{-1}(V) \rightarrow Y - V$  is bijective,  $f : X - f^{-1}(V) \rightarrow Y - V$  is biholomorphic. Additionally,  $f$  is surjective by the definition of a bimeromorphic morphism. Consequently, there exists an irreducible component  $C_t$  of  $X_t$  such that  $f(C_t) = D_t$  by the irreducibility of  $D_t$ .

In view of the codimensions of  $V$  and  $D_t$ , it follows that  $D_t \not\subseteq V$ , and consequently,  $C_t \not\subseteq f^{-1}(V)$ . Clearly,  $D_t \cap V$  is a thin analytic subset of  $D_t$ , and  $C_t \cap f^{-1}(V)$  is a thin analytic subset of  $C_t$ . Hence, one can easily check by definition that  $f : C_t \rightarrow D_t$  is bimeromorphic.  $\square$

We next prove a simplified version of the fiberwise bimeromorphic criterion using plurigenera; for a much more general version, please refer to [CRT25].

**Theorem 4.3** ([CRT25, Theorem 1.6]). Let

$$\pi_1 : X \rightarrow S, \pi_2 : Y \rightarrow S$$

be two (locally) Moishezon morphisms with irreducible fibers that admit canonical singularities, such that  $\kappa(X_0) \geq 0$ . Then the bimeromorphic map that connects  $\pi_1$  and  $\pi_2$  is indeed fiberwise bimeromorphic.

Let us briefly sketch out the idea. We first take the resolution of indeterminacy; by further resolution we can guarantee that the generic fibers of  $W \rightarrow S$  are smooth.

$$\begin{array}{ccc} & W & \\ p \swarrow & & \searrow q \\ X & \dashrightarrow & Y \\ \pi_1 \searrow & & \swarrow \pi_2 \\ & S & \end{array}$$

We claim that the strict transform  $\tilde{X}_0 = p_*^{-1}(X_0)$  and the strict transform  $\tilde{Y}_0 = q_*^{-1}(Y_0)$  must coincide. Otherwise, since plurigena are bimeromorphic invariants, we have  $P_m(X_0) = P_m(\tilde{X}_0)$ ,  $P_m(Y_0) = P_m(\tilde{Y}_0)$  and  $P_m(W_t) = P_m(X_t) = P_m(Y_t)$ . On the other hand, since the family  $W \rightarrow S$  is Moishezon, by the lower semi-continuity of the plurigena (for Moishezon morphisms), we have  $P_m(\tilde{X}_0) + P_m(\tilde{Y}_0) \leq P_m(W_t)$ . Since  $Y \rightarrow S$  is a Moishezon morphism with fiberwise canonical singularities, the plurigena remain constant, i.e.  $P_m(Y_t) = P_m(Y_0)$ . Putting those together, we have

$$P_m(X_0) + P_m(\tilde{Y}_0) = P_m(\tilde{X}_0) + P_m(\tilde{Y}_0) \leq P_m(W_t) = P_m(X_t) = P_m(Y_t) = P_m(Y_0) = P_m(\tilde{Y}_0)$$

so that the plurigenus  $P_m(X_0) = 0$ , which contradicts  $\kappa(X_0) \geq 0$ .

## 5 The Fiberwise Bimeromorphic Conjecture for Moishezon Morphisms

In the last section, we will prove the following conjecture under the additional assumption that the center fiber is KLT and not uniruled.

**Conjecture 5.1** (Fiberwise bimeromorphic conjecture for Moishezon morphism, [Kol22, Conjecture 5]). Let  $g : X \rightarrow \mathbb{D}$  be a flat, proper, Moishezon morphism. Assume that  $X_0$  has canonical (resp. log terminal) singularities.

Then  $g$  is fiberwise birational to a flat, projective morphism  $g^p : X^p \rightarrow \mathbb{D}$  such that

- (1)  $X_0^p$  has canonical (resp. log terminal) singularities,
- (2)  $X_s^p$  has terminal singularities for  $s \neq 0$ , and
- (3)  $K_{X^p}$  is  $\mathbb{Q}$ -Cartier.

**Remark 5.2.** Before continuing our discussion of this conjecture, let us first look closely at what this conjecture is about. The conjecture shows that the flat Moishezon morphism is not only bimeromorphic to some projective model but it is indeed fiberwise bimeromorphic to some projective model, as long as the singularity on the central fiber is nice enough.

Kollár verifies the conjecture when the central fiber is KLT and not uniruled. Before proving the theorem, let us list some intermediate results that will be used.

**Proposition 5.3** (Inversion of adjunction, [Kol22, Proposition 30]). Let  $X$  be a normal complex analytic space,  $X_0 \subset X$  a Cartier divisor, and  $\Delta$  an effective  $\mathbb{R}$ -divisor such that  $K_X + \Delta$  is  $\mathbb{R}$ -Cartier. Then  $(X, X_0 + \Delta)$  is PLT in a neighborhood of  $X_0$  iff  $(X_0, \Delta|_{X_0})$  is KLT.

**Proposition 5.4** (Existence of canonical modification, [Kol22, Corollary 38]). Let  $f : X \rightarrow \mathbb{D}$  be a flat, proper, Moishezon morphism. Assume that  $X_0$  is log terminal. Then  $X$  has a canonical modification  $\pi : X^c \rightarrow X$ , such that

- (a)  $X_0^c$  is log terminal and,
- (b)  $\pi$  is fiberwise birational.

*Proof.* The proof uses some algebraic approximation technique, see [Kol22]. □

**Lemma 5.5** ([Kol22, Lemma 31.1]). Let  $X \rightarrow S$  be a proper, Moishezon morphism,  $D$  an  $\mathbb{R}$ -divisor on  $X$ , and  $A$  a big  $\mathbb{R}$ -divisor on  $X$  such that  $\mathbf{B}^{\text{div}}(A) = \emptyset$ . Then, for every prime divisor  $F \subset X$ ,

$$\text{coeff}_F \mathbf{B}_-^{\text{div}}(D) = \lim_{\epsilon \rightarrow 0} \text{coeff}_F \mathbf{B}_-^{\text{div}}(D + \epsilon A)$$

**Lemma 5.6** ([Kol22, Lemma 31.2]). Let  $X_i \rightarrow S$  be proper, Moishezon morphisms,  $h : X_1 \rightarrow X_2$  a proper, bimeromorphic morphism,  $D_2$  a pseudo-effective,  $\mathbb{R}$ -Cartier divisor on  $X_2$ , and  $E$  an effective,  $h$ -exceptional divisor. Then

$$\mathbf{B}_-^{\text{div}}(E + h^* D_2) \geq E.$$

The following proposition is useful in the proof.

**Proposition 5.7.** Let  $f : X \rightarrow U$  be a proper morphism between complex varieties,  $(X, \Delta)$  a DLT pair and  $\phi : X \dashrightarrow X_M$  be a minimal model for  $K_X + \Delta$  over  $U$ . Then the set of  $\phi$ -exceptional divisors coincides with the set of divisors contained in  $\mathbf{B}_-(K_X + \Delta/U)$ .

*Proof.* Let  $p : Y \rightarrow X$  and  $q : Y \rightarrow X_M$  be a common resolution. Since  $\phi$  is  $(K_X + \Delta)$ -negative, we have that  $p^*(K_X + \Delta) = q^*(K_{X_M} + \phi_* \Delta) + E$  where  $E$  is effective,  $q$ -exceptional and the support of  $p_* E$  is the set of  $\phi$ -exceptional divisors. Since the minimal model assumption gives  $N_\sigma(p^*(K_X + \Delta)/U) = E$ , we get

$$p_* E = N_\sigma(K_X + \Delta).$$

□

**Lemma 5.8.** Let  $b_0 = 1, b_1, \dots, b_n$  be real numbers which are linearly independent over  $\mathbb{Q}$ , and suppose that the divisor  $\sum_{i=0}^n b_i B_i$  is  $\mathbb{R}$ -Cartier. Then each of the divisors  $B_i$  is  $\mathbb{Q}$ -Cartier.

Having introduced a number of lemmas that will be used in the proof, we can now dive into the proof of the last main theorem of this note.

**Theorem 5.9** (A flat Moishezon morphism with KLT and non-uniruled central fiber will be fiberwise bimeromorphic to a projective morphism, [Kol22], Theorem 28). Let  $g : X \rightarrow \mathbb{D}$  be a flat, proper, Moishezon morphism. Assume that

1.  $X_0$  has log terminal singularities and
2.  $X_0$  is not uniruled

Then

- (a)  $g$  is fiberwise birational to a flat, projective morphism  $g^p : X^p \rightarrow \mathbb{D}$  (possibly over a smaller disc),
- (b)  $X_0^p$  has log terminal singularities,
- (c)  $X_s^p$  is not uniruled and has terminal singularities for  $s \neq 0$ ,
- (d)  $K_{X^p}$  is  $\mathbb{Q}$ -Cartier

*Proof.* We take a resolution of singularities  $Y \rightarrow X$  such that  $Y \rightarrow \mathbb{D}$  is projective, and then take a relative minimal model of  $Y \rightarrow \mathbb{D}$ . We hope that it gives what we want. There are, however, several obstacles.

**Step 1. Take the canonical modification.** We need to control the singularities of  $X$ . First, for a flat proper Moishezon morphism with KLT central fiber, there exists a canonical modification (Theorem 5.4) which is fiberwise birational and the central fiber is KLT. Thus we are in the case that  $K_X$  is  $\mathbb{Q}$ -Cartier.

Indeed, by the canonical modification we can find some canonical modification  $X^c \rightarrow X$  such that  $X^c$  has canonical singularities and the morphism  $X^c \rightarrow X$  is fiberwise birational. Thus, if we can prove the result for  $X^c \rightarrow \mathbb{D}$  then it will also be true for  $X \rightarrow \mathbb{D}$  (since the composition of fiberwise birational maps is again fiberwise birational).

We assume this from now on. Then the inversion of adjunction for PLT pairs implies that the pair  $(X, X_0)$  is PLT, by setting  $\Delta = 0$  in the inversion of adjunction. (To apply the inversion of adjunction here we require  $K_X$  to be  $\mathbb{Q}$ -Cartier.)

**Step 2. Take the semi-stable reduction.** After a base change  $z \mapsto z^r$  we get  $g^r : X^r \rightarrow \mathbb{D}$ . For suitable  $r$ , there is a semi-stable, projective resolution  $h : Y \rightarrow \mathbb{D}$ ; we may also choose it to be equivariant for the action of the cyclic group  $G \cong \mathbb{Z}_r$ . All subsequent steps will be  $G$ -equivariant. We denote by  $X_0^Y$  the birational transform of  $X_0$  and by  $E_i$  the other irreducible components of  $Y_0$ .

$$\begin{array}{ccccc}
 Y & \longrightarrow & X^r & \longrightarrow & X \\
 & \searrow & \downarrow & & \downarrow \\
 & & \mathbb{D} & \longrightarrow & \mathbb{D}
 \end{array}$$

Such that the following conditions hold:

- (a)  $Y$  is non-singular,

- (b) generic fibers are non-singular,
- (c) The special fiber is a reduced divisor with SNC support,
- (d) Denote  $Y_0 = X_0^Y + \sum c_i E_i$  (with  $X_0^Y$  being the strict transform of  $X_0$ ); note that the strict transform  $X_0^Y$  dominates  $X_0$ .

**Step 3. Prove the generic fibers  $Y_s$  are not uniruled (for  $s \neq 0$ ).** We will prove it by contradiction: suppose the generic fibers  $Y_s$  are uniruled. Then, by Matsusaka's theorem (see [Kol96, Theorem VI.1.7]), all the irreducible components of  $Y_0$  are uniruled. On the other hand, since  $X_0^Y$  dominates  $X_0$ ,  $X_0$  must be uniruled, a contradiction.

Finally, by the BDPP theorem, it is easy to see that  $K_{Y_s}$  is pseudo-effective (since we assume that generic fibers are smooth).

**Step 4. Run the MMP.** We require the condition that the general fibers are of log general type. To achieve this, let  $H$  be an ample,  $G$ -equivariant divisor such that  $Y_0 + H$  is snc (note that this is possible by taking  $H' = \otimes_{m=1}^n g \cdot H$ ; since  $G$  is a finite group, this is a well-defined ample line bundle). For  $\epsilon > 0$  we get a pair  $(Y, \epsilon H)$  whose general fibers  $(Y_s, \epsilon H_s)$  are of log general type since  $K_{Y_s}$  is pseudo-effective by the previous step. For such algebraic families, relative minimal models are known to exist by BCHM. We also know that  $(Y, Y_0 + \epsilon H)$  is dlt for  $0 < \epsilon \ll 1$  (since  $Y$  is smooth and  $Y_0 + H$  is snc).

Thus we get the  $(K_X + Y_0 + \epsilon H)$ -relative MMP on the disc  $\mathbb{D}$ , (note that the base is an analytic disc, thus the MMP is in the sense of Fujino [Fuj22] or Kollár-Nicaise-Xu [KNX18]).

$$\begin{array}{ccc} (Y, \epsilon H) & \xrightarrow{\phi} & (Y^m, \epsilon H^m) \\ & \searrow & \swarrow \\ & \mathbb{D} & \end{array}$$

We claim  $(Y^m, Y_0^m + \epsilon H^m)$  is DLT, and  $H^m$  is  $\mathbb{Q}$ -Cartier for a general choice of  $\epsilon$ , and thus  $(Y^m, Y_0^m)$  is also dlt.

Indeed, each step of the MMP will preserve the DLT condition (see [BCHM] Lemma 3.10.10), so it is easy to see that  $(Y^m, Y_0^m + \epsilon H^m)$  is DLT. On the other hand, by Lemma 5.8, it is easy to see that if  $\epsilon$  is sufficiently general, the  $\mathbb{Q}$ -linear independence condition is satisfied, and therefore  $H^m$  is indeed a  $\mathbb{Q}$ -Cartier divisor. And finally by [KM98, Corollary 2.39]  $(Y^m, Y_0^m)$  is also DLT.

Recall that we have

$$\mathbf{B}_-^{\text{div}}(K_Y + Y_0) \geq (1 + a(E_i, X^r, X_0))E_i,$$

since the discrepancy of a PLT pair  $a(E_i, X^r, X_0) > -1$ , thus all the exceptional divisors  $E_i$  are contained in the divisorial part of the restricted base locus  $\mathbf{B}_-^{\text{div}}(K_Y + Y_0)$ . On the other hand

$$\text{coeff}_F \mathbf{B}_-^{\text{div}}(D) = \lim_{\epsilon \rightarrow 0} \text{coeff}_F \mathbf{B}_-^{\text{div}}(D + \epsilon A),$$

for any prime divisor  $F$ . Thus, for sufficiently small  $\epsilon$ ,  $E_i$  is also contained in the restricted base locus of  $K_Y + Y_0 + \epsilon H$  (since the coefficient of  $E_i$  in  $\mathbf{B}_-^{\text{div}}(K_Y + Y_0 + \epsilon H)$  is also positive if  $\text{coeff}_{E_i} \mathbf{B}_-^{\text{div}}(K_Y + Y_0) > 0$ ). Then, by Proposition 5.7, any MMP will contract those  $E_i$ .

**Step 5. Prove fiberwise bimeromorphic.** By the Cone theorem, those divisors contracted will be covered by rational curves. However, we assume that  $X_0^Y$  is not uniruled (thus, it is not

contracted by the MMP). By Theorem 2.4 the generic fiber of  $X \dashrightarrow Y^m$  is bimeromorphic, so one only needs to prove that the central fiber  $X_0$  is bimeromorphic to  $Y_0^m$ . In fact, since the only component of  $Y_0^m$  is the strict transform of  $X_0^Y$ ,  $X_0$  is bimeromorphic to  $Y_0^m$ .

**Step 6. Check the singularity assumptions.** Note that the fibers  $Y_s$  of the family  $h : Y \rightarrow \mathbb{D}$  are smooth away from  $Y_0$  (by the semi-stable assumption), thus  $(Y_s, \epsilon H_s)$  is terminal for  $s \neq 0$  and  $0 \leq \epsilon \ll 1$  (see [KM98, Corollary 2.35. (2)]).

Since  $H_s$  is ample, by the negativity lemma the MMP above will not contract  $H_s$ . Note that  $(Y_s^m, \epsilon H_s^m)$  is still terminal (by [KM98, Corollary 3.43]). Thus,  $Y_s^m$  also admits terminal singularities (see [KM98, Corollary 2.35]). Since  $(Y^m, Y_0^m)$  is DLT, it is also PLT thanks to the irreducibility of  $Y_0^m$  ([KM98, Proposition 5.51]). And therefore  $Y_0^m$  is KLT by the easy direction of the inversion of adjunction (see Theorem 5.3).  $\square$

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## 1 Overview

The aim of this note is to study the distribution of the general type locus, the Moishezon locus, and the projective locus (see Definition 2.1) on the irreducible base. The motivation for this topic comes from the following observation on the distribution of polarized (projective) K3 surfaces in the universal family of marked complex K3 surfaces.

Let  $X \rightarrow D^{20}$  be a universal family of K3 surfaces. A smooth, compact surface is Moishezon iff it is projective. The projective fibers of  $X \rightarrow D^{20}$  correspond to a countable union of hypersurfaces  $H_{2g} \subset D^{20}$ . As we can see from this example, the projective locus (which corresponds to projective K3 surfaces) is a countable union of hypersurfaces in the moduli space  $D^{20}$ .

It is natural to ask how the locus of fibers that admit a certain property is distributed on the base. This note focuses on the distribution of the fibers that are projective, of general type, and Moishezon on the base  $S$ . The main references of this note are [Kol22a], and [Kol22b], [RT22].

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## 2 The alternating property of the very big locus, general type locus

We first give the definitions of the very big locus, Moishezon locus, general type locus, and the projective locus.

**Definition 2.1** (Very big locus, general type locus, Moishezon locus, [Kol22a, Definition 18]).

Let  $g : X \rightarrow S$  be a proper morphism of normal analytic spaces and  $L$  a line bundle on  $X$ . Set

1.  $\text{VB}_S(L) := \{s \in S : L_s \text{ is very big on } X_s\} \subset S$ ,



2.  $\text{GT}_S(X) := \{s \in S : X_s \text{ is of general type}\} \subset S$ ,
3.  $\text{MO}_S(X) := \{s \in S : X_s \text{ is Moishezon}\} \subset S$ ,
4.  $\text{PR}_S(X) := \{s \in S : X_s \text{ is projective}\} \subset S$ .

Here very big means the locus  $s \in S$  such that  $X_s \dashrightarrow \text{Proj}_S(g_*L_s)$  is bimeromorphic onto the closure of its image.

**Definition 2.2** (Locus  $V$  that satisfies the alternating property over  $S$ ). Let  $g : X \rightarrow S$  be a proper morphism of normal analytic spaces. We say the locus

$$V := \{s \in S \mid X_s \text{ admits property } P\},$$

satisfies *the alternating property over  $S$*  if  $V \subset S$  is

- (1) either nowhere dense (in the analytic Zariski topology),
- (2) or it contains a dense open subset of  $S$ .

**Remark 2.3.** In general: (a) A subset which is not nowhere dense need not contain an open subset of  $S$ , e.g.  $\mathbb{Q} \subset \mathbb{R}$  is not nowhere dense but it clearly contains no dense open subset of  $\mathbb{R}$ . (b) A subset that is not nowhere dense need not contain a dense subset of  $S$  either, e.g. the disc  $\mathbb{D} = \{|z| < 1\} \subset \mathbb{C}$  is not nowhere dense, but it is also not dense in  $\mathbb{C}$ .

In the analytic Zariski topology: if  $S$  is irreducible and  $V \subset S$  is not nowhere dense, then  $V$  is dense in  $S$ . To see this, by definition,  $\bar{V}$  contains a non-empty Zariski open subset of  $S$ . Since  $S$  is irreducible, every non-empty Zariski open subset is dense and therefore  $\bar{V} = S$ .

Note that whether  $V \subset S$  satisfies the alternating property over  $S$  does not depend on information about the special fibers. In other words, we have the following lemma.

**Lemma 2.4.** Assume that  $S$  is irreducible. If  $V$  satisfies the alternating property on some non-empty Zariski open subset  $S' \subset S$ , then  $V$  also satisfies the alternating property on  $S$ .

*Proof.* Since  $S$  is irreducible, the non-empty Zariski open subset  $S' \subset S$  is Zariski dense in  $S$ . Then we have two cases:

Case 1. If  $V$  is nowhere dense in  $S'$ , then  $V$  is also nowhere dense in  $S$ . By contradiction, suppose there exists some non-empty Zariski open subset  $W$  (of  $S$ ) contained in  $V$ . Since  $S$  is irreducible, the intersection  $W \cap S'$  is a non-empty Zariski open subset of  $S'$ , and therefore it contradicts the nowhere density of  $V$  in  $S'$ .

Case 2. If  $V$  is dense in  $S'$  and we know that  $S' \subset S$  is Zariski dense, then  $V$  is also dense in  $S$ .

□

Note that a local system on an irreducible complex variety is trivial.

**Lemma 2.5.** Local systems on an irreducible algebraic variety with the Zariski topology are trivial.

*Proof.* Since  $X$  is irreducible iff any non-empty intersection of Zariski open subsets is non-empty. And by definition, for every point  $x \in X$ , there exists an open subset on which the local system is constant,

$$\mathcal{L}|_{U_i} = \underline{S}_i,$$

and every such  $U_i \cap U_j \neq \emptyset$  satisfies

$$\mathcal{L}|_{U_i \cap U_j} = \underline{S}_i|_{U_i \cap U_j} = \underline{S}_j|_{U_i \cap U_j},$$

so that  $\mathcal{L}$  is constant on  $U_i \cup U_j$ . By quasi-compactness, we know that the local system is automatically constant.  $\square$

We first show that the very big locus satisfies the alternating property.

**Proposition 2.6.** Let  $f : X \dashrightarrow Y/S$  be a proper morphism between complex analytic varieties. Assume that the restriction on each fiber  $f_s : X_s \dashrightarrow Y_s$  is bimeromorphic. Can we prove that  $f : X \rightarrow Y/S$  is a bimeromorphic  $S$ -map?

**Theorem 2.7** (Alternating property of very big locus, [Kol22a, Lemma 19]).

Let  $g : X \rightarrow S$  be a proper morphism of normal irreducible analytic spaces and  $L$  a line bundle on  $X$ . Then  $\text{VB}_S(L) \subset S$  is

- (1) either nowhere dense (in the analytic Zariski topology),
- (2) or it contains a dense open subset of  $S$ , and  $g : X \rightarrow S$  is Moishezon.

*Proof.* We may assume that  $g : X \rightarrow S$  is surjective (otherwise, by properness of  $g$ , it will immediately fall in (1)). By Lemma 2.4, it is possible to pass to a non-empty Zariski open subset of  $S$ . Thus, we may assume that  $g$  is flat,  $g_*L$  is locally free, and commutes with restriction to the fibers. We get a meromorphic map  $\phi : X \dashrightarrow \mathbb{P}_S(g_*L)$ . There is thus a smooth, bimeromorphic model  $\pi : X' \rightarrow X$  such that  $\phi \circ \pi : X' \rightarrow \mathbb{P}_S(g_*L)$  is a morphism. After replacing  $X$  by  $X'$  and again passing to an open subset of  $S$ , we may assume that  $g$  is flat,  $g_*L$  is locally free, commutes with restriction to fibers, and  $\phi : X \rightarrow \mathbb{P}_S(g_*L)$  is a morphism.

Let  $Y \subset \mathbb{P}_S(g_*L)$  denote its image and  $W \subset X$  the Zariski closed set of points where  $\pi : X \rightarrow Y$  is not smooth. Set  $Y^\circ := Y \setminus \phi(W)$  and  $X^\circ := X \setminus \phi^{-1}(\phi(W))$ . The restriction  $\phi^\circ : X^\circ \rightarrow Y^\circ$  is then smooth and proper. We divide the discussion into two cases:

Case 1. Assume that the set of points

$$E = \{y \in Y \mid \phi^{-1}(y) \text{ is a single point}\} \subset Y,$$

is not dense in  $Y$ . We claim that in this case  $\text{VB}_S(L)$  is nowhere dense in  $S$ . For otherwise, it would imply that  $\text{VS}_S(L)$  is dense in  $S$ , and so that for a dense set of fibers  $\{X_s\}_{s \in \text{VB}_S(L)} \subset X$ , the restriction of the relative Kodaira map is bimeromorphic onto its image. In particular,  $E_s = E \cap X_s \subset X_s$  is dense in  $X_s$ . We claim that this implies that

$$E = \bigcup_{s \in S} E_s,$$

is dense in  $Y$ , which gives the contradiction. This is because

$$\bigcup_{s \in \text{VB}_S(L)} Y_s = \bigcup_{s \in \text{VB}_S(L)} \text{cl}_{X_s}(E_s) \subset \bigcup_{s \in \text{VB}_S(L)} \text{cl}_X(E_s) \subset \text{cl}_X\left(\bigcup_{s \in \text{VB}_S(L)} E_s\right)$$

Case 2.  $E$  is a dense set in  $Y$ , thus it is also dense in  $Y^\circ$ . Since  $\phi^\circ$  is proper and smooth,  $\phi^\circ$  is a finite étale morphism of degree 1, thus it is an isomorphism.

Thus,  $\phi$  is bimeromorphic on every irreducible fiber that has a non-empty intersection with  $X^\circ$ . That is, if we denote  $D := \{s \in S \mid X_s \cap X^\circ \neq \emptyset\} \cap \{s \in S \mid X_s \text{ is irreducible}\}$  with  $g(X^\circ) = \{s \in S \mid X_s \cap X^\circ \neq \emptyset\}$ , then

$$D \subset \text{VB}_S(L),$$

(1) Note that irreducibility of the fiber  $X_s$  is needed: if  $X_s \cap X^\circ \neq \emptyset$  and  $X_s$  is irreducible, then  $X^\circ \cap X_s \subset X_s$  is a non-empty Zariski open subset of  $X_s$ , which is dense in the fiber  $X_s$ . Note again that since both  $X$  and  $S$  are irreducible, the generic fibers of  $g$  are irreducible, see [GW20, Exercise 6.15]. Thus adding this constraint will not change the result,

(2) Note that the very big locus is not directly defined by the restriction of  $X \rightarrow \mathbb{P}_S(g_*L)$  on the fibers. Instead, it is defined by the Kodaira map  $X_s \rightarrow \mathbb{P}(H^0(X_s, L_s))$ . Since we assume that  $g_*L$  commutes with restriction on the fiber, these two Kodaira maps coincide.

Recall that a morphism between analytic varieties sends a dense subset to a dense subset in its image. And  $g$  is flat (by assumption at the beginning), so  $g$  is open. Thus  $g$  sends a Zariski dense open subset to a Zariski dense open subset. Thus  $D$  is a non-empty dense Zariski open subset contained in  $\text{VB}_S(L)$ .

Finally, we need to show that in this case  $g : X \rightarrow S$  is a Moishezon morphism, i.e. the relative Kodaira map over  $S$  induced by  $L$  is bimeromorphic onto its image. Since  $\phi^\circ : X^\circ \rightarrow Y^\circ$  is an isomorphism for  $X^\circ \subset X$  a non-empty dense open subset, the result follows.  $\square$

As a direct consequence (combined with the birational boundedness result of Hacon–McKernan [HM06]) we see that the general type locus also admits the alternating property.

**Theorem 2.8** (The alternating property of the general type locus, [Kol22a, Corollary 20]).

Let  $g : X \rightarrow S$  be a proper morphism of normal, irreducible analytic spaces. Then the general type locus

$$\text{GT}_S(X) = \{s \in S \mid X_s \text{ is of general type}\},$$

- (1) either nowhere dense (in the analytic Zariski topology),
- (2) or it contains a dense open subset of  $S$ , and  $g : X \rightarrow S$  is Moishezon

*Proof.* Using resolution of singularities, we may assume that  $X$  is smooth. By passing to an open subset of  $S$ , we may also assume that  $S$  and  $g$  are smooth. By [HM06] there is an  $m$  (depending only on  $\dim X_s$ ) such that  $|mK_{X_s}|$  is very big whenever  $X_s$  is of general type. Thus, Theorem 2.7 applies to  $L = mK_X$ .  $\square$

### 3 The alternating property of the Moishezon locus

In this section, we will prove that the Moishezon locus also admits a certain alternating property. Before proving Theorem 3.6, let us first introduce the following result, due to [RT22].

**Definition 3.1.** Let  $\mathcal{X}$  be a complex manifold,  $\Delta \subseteq \mathbb{C}$  the unit disk, and  $f : \mathcal{X} \rightarrow \Delta$  a flat family, smooth over the punctured disk  $\Delta^*$ . We say that  $f$  is a one-parameter degeneration.

**Theorem 3.2** (Moishezon morphism criterion, [RT22, Proposition 3.15]). Let  $\pi : \mathcal{X} \rightarrow \Delta$  be a smooth morphism.

(1) Assume that there exists an uncountable subset  $B$  of  $\Delta$  such that for each  $t \in B$ , the fiber  $X_t$  admits a line bundle  $L_t$  with the property that  $c_1(L_t)$  comes from the restriction to  $X_t$  of some cohomology class in  $H^2(\mathcal{X}, \mathbb{Z})$ .

(2) Assume further that the Hodge number  $h^{0,2}(X_t) := h^2(X_t, \mathcal{O}_{X_t})$  is independent of  $t \in \Delta$  (the original theorem requires only Hodge (0,1) deformation invariance).

Then there exists a global line bundle  $L$  over  $\mathcal{X}$  such that  $c_1(L|_{X_s}) = c_1(L_s)$  for some  $s$  in some uncountable subset of  $B$ .

*Proof.* Apply the sheaf exponential exact sequence so that

$$0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O}_{\mathcal{X}} \rightarrow \mathcal{O}_{\mathcal{X}}^* \rightarrow 0.$$

We claim that

$$H^2(\mathcal{X}, \mathcal{O}_{\mathcal{X}}) \cong H^0(\Delta, R^2\pi_*\mathcal{O}_{\mathcal{X}}), \quad H^2(X_s, \mathcal{O}_{X_s}) \cong R^2\pi_*\mathcal{O}_{\mathcal{X}}(s).$$

Indeed:

(1) By Cartan B, we have

$$H^p(S, R^q\pi_*\mathcal{O}_{\mathcal{X}}) = 0, \quad p > 0, q \geq 0,$$

and the Leray spectral sequence argument implies the first one,

(2) Since we assume the cohomological dimension  $h^{0,2}$  is constant, by the Grauert base change theorem, the second one follows.

Thus, we have the following commutative diagram.

$$\begin{array}{ccccccc}
 & & & & H^0(\mathcal{X}, R^2\pi_*\mathcal{O}_{\mathcal{X}}) & & \\
 & & & & \downarrow \cong & & \\
 \longrightarrow & H^1(\mathcal{X}, \mathcal{O}_{\mathcal{X}}^*) & \longrightarrow & H^2(\mathcal{X}, \mathbb{Z}) & \xrightarrow{e_2} & H^2(\mathcal{X}, \mathcal{O}_{\mathcal{X}}) & \longrightarrow \\
 & \downarrow & & \downarrow & & \downarrow & \\
 \longrightarrow & H^1(X_s, \mathcal{O}_{X_s}^*) & \longrightarrow & H^2(X_s, \mathbb{Z}) & \xrightarrow{e_2} & H^2(X_s, \mathcal{O}_{X_s}) & \longrightarrow \\
 & & & & & \downarrow \cong & \\
 & & & & & R^2\pi_*\mathcal{O}_{\mathcal{X}}(s) & 
 \end{array}$$

Where we have the evaluation  $\text{ev}_s : H^0(\mathcal{X}, R^2\pi_*\mathcal{O}_{\mathcal{X}}) \rightarrow R^2\pi_*\mathcal{O}_{\mathcal{X}}(s)$  in the diagram above.

Let  $L_s \in \text{Pic}(X_s)$  be such that  $c_1(L_s) \in H^2(X_s, \mathbb{Z})$ . By simple connectedness of  $\Delta$ ,  $c_1(L_s) \in H^2(X_s, \mathbb{Z})$  lifts to  $c \in H^2(\mathcal{X}, \mathbb{Z})$ . If we can prove the vanishing of  $e_2(c) \in H^2(\mathcal{X}, \mathcal{O}_{\mathcal{X}})$ , then by the exactness of the sequence we can find some global line bundle  $L \in \text{Pic}(\mathcal{X})$ .

Observe that the cohomology group  $H^2(X_s, \mathbb{Z}) \cong H^2(\mathcal{X}, \mathbb{Z})$  has  $\mathbb{Z}$  coefficients, so that it has only countably many elements. Given uncountably many  $L_t$ , there must exist some  $c \in H^2(\mathcal{X}, \mathbb{Z})$  such that an uncountable subset of  $t$  satisfies  $c_1(L_t) = c$ .

Since this  $c \in H^2(X_s, \mathbb{Z})$  comes from  $L_s \in \text{Pic}(X_s)$ , we have  $e_2(c) = 0 \in R^2\pi_*\mathcal{O}_{\mathcal{X}}(s)$ , and thus, if we lift it to  $c \in H^2(\mathcal{X}, \mathbb{Z})$ , the global section  $e_2(c) \in H^0(\mathcal{X}, R^2\pi_*\mathcal{O}_{\mathcal{X}})$  will vanish on uncountably many points. Thus, by the identity principle (since  $R^2\pi_*\mathcal{O}_{\mathcal{X}}$  is locally free (this step is not due to torsion freeness; we need the Hodge number condition to get torsion freeness) the vanishing locus of  $e_2(c)$  is a subvariety), we have  $e_2(c) = 0 \in H^2(\mathcal{X}, R^2\pi_*\mathcal{O}_{\mathcal{X}})$ . Therefore, there exists a global line bundle  $L \in \text{Pic}(\mathcal{X})$  with the restriction  $c_1(L|_{X_s}) = c_1(L_s)$ . Finally, by Proposition 3.3 on deformation density of the Iitaka-Kodaira dimension, we conclude that  $L$  is indeed a global big line bundle.  $\square$

The following proposition is used in the proof of Theorem 3.1.

**Proposition 3.3** (Deformation density of Iitaka-Kodaira dimension, [LS77, Theorem 3.4]). Let  $\pi : \mathcal{X} \rightarrow Y$  be a flat family from a complex manifold over a one-dimensional connected complex manifold  $Y$  with possibly reducible fibers. If there exists a holomorphic line bundle  $L$  on  $\mathcal{X}$  such that the Kodaira-Iitaka dimension  $\kappa(L_t) = \kappa$  for each  $t$  in an uncountable set  $B$  of  $Y$ , then any fiber  $X_t$  in  $\pi$  has at least one irreducible component  $C_t$  with  $\kappa(L|_{C_t}) \geq \kappa$ .

In particular, if any fiber  $X_t$  for  $t \in Y$  is irreducible, then  $\kappa(L_t) \geq \kappa$ .

We next add some supplementary material about the sheaf exponential sequence and relative Picard functor.

**Lemma 3.4** ([Har77, p. 466]). Let  $X$  be a reduced complex analytic space. Then the following sheaf exponential sequence is exact.

$$0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_X^* \rightarrow 0.$$

The following proposition concerns the relative Picard functor.

**Proposition 3.5** (Picard-Brauer exact sequence). Let  $X \rightarrow T$  be a proper surjective morphism between complex varieties.

$$\begin{aligned} 0 \rightarrow H^1(T, f_{T*}\mathcal{O}_{X_T}^*) &\rightarrow H^1(X_T, \mathcal{O}_{X_T}^*) \rightarrow H^0(T, R^1f_{T*}\mathcal{O}_{X_T}^*) \\ &\rightarrow H^2(T, f_{T*}\mathcal{O}_{X_T}^*) \rightarrow H^2(X_T, \mathcal{O}_{X_T}^*), \end{aligned}$$

we call  $H^2(T, f_{T*}\mathcal{O}_{X_T}^*) = H^2(T, \mathcal{O}_T^*)$  the Brauer group. Note that a global section of  $H^0(T, R^1(f_T)_*\mathcal{O}_{X_T}^*)$  that comes from  $H^0(X_T, R^2(f_T)_*\mathbb{Z})$  will automatically vanish in the Brauer group, since we have the factorization of  $H^2(T, \mathcal{O}_{X_T}^*)$  as

$$H^0(T, R^1(f_T)_*\mathcal{O}_{X_T}^*) \rightarrow H^0(T, R^2(f_T)_*\mathbb{Z}_{X_T}) \rightarrow H^0(T, R^2(f_*)).$$

If  $\mathcal{O}_S \xrightarrow{\sim} f_*\mathcal{O}_X$  holds universally, then  $H^1(T, \mathcal{O}_T^*) \xrightarrow{\sim} H^1(T, f_{T*}\mathcal{O}_{X_T}^*)$ . Hence we have the following Picard-Brauer exact sequence,

$$0 \rightarrow \text{Pic}(T) \rightarrow \text{Pic}(X_T) \rightarrow \text{Pic}_{(X/S)(\text{zar})}(T).$$

We now turn to the proof of the alternating property of the Moishezon locus.

**Theorem 3.6** ([Kol22a, Theorem 21]). Let  $g : X \rightarrow S$  be a smooth, proper morphism of normal, irreducible analytic spaces. Then  $\text{MO}_S(X) \subset S$  is

- (1) either contained in a countable union  $\cup_i Z_i$ , where  $Z_i \subsetneq S$  are Zariski closed,
- (2) or  $\text{MO}_S(X)$  contains a dense, open subset of  $S$ .

Furthermore, if  $R^2 g_* \mathcal{O}_X$  is torsion free, then (2) can be replaced by

- (3)  $\text{MO}_S(X) = S$  and  $g$  is locally Moishezon.

**Remark 3.7.** Condition (1) is slightly different from the nowhere dense condition compared with Lemma 2.7 and Theorem 2.8. Indeed, a countable union of nowhere dense subsets need not be nowhere dense (e.g.  $\mathbb{Q}$ , as a countable union of nowhere dense subsets, is no longer nowhere dense). As we will see in the proof, this replacement is necessary. Another difference compared with Lemma 2.7 and Proposition 2.8 is that here we assume the morphism is smooth.

**Remark 3.8.** As mentioned by Professor Kollár, the proof is motivated by [RT22]. Compared with the proof of [RT22], Kollár's proof does not require the base to be the unit disc  $\Delta$ . Consequently, the direct image  $R^2 g_* \mathcal{O}_X$  is only torsion free, which need not be locally free.

*Proof.* Assume first that  $R^2 g_* \mathcal{O}_X$  is torsion free. The sheaf exponential sequence

$$0 \rightarrow \mathbb{Z}_X \rightarrow \mathcal{O}_X \xrightarrow{\exp} \mathcal{O}_X^* \rightarrow 1.$$

gives

$$R^1 g_* \mathcal{O}_X^* \rightarrow R^2 g_* \mathbb{Z}_X \xrightarrow{e_2} R^2 g_* \mathcal{O}_X.$$

We may pass to the universal cover of  $S$ . Note that the local system on the simply connected space is constant, thus  $R^2 g_* \mathbb{Z}_X \otimes \mathcal{O}_S$  is a trivial bundle.

Let  $\{\ell_i\}$  be those global sections of  $R^2 g_* \mathbb{Z}_X$  such that  $e_2(\ell_i) \in H^0(S, R^2 g_* \mathcal{O}_X)$  is identically 0, and  $\{\ell'_j\}$  the other global sections (these  $\{\ell_i, \ell'_j\}$  are countable since we consider the  $\mathbb{Z}$ -coefficient local system). The  $\ell_i$  then lift back to global sections of  $R^1 g_* \mathcal{O}_X^*$ , hence to line bundles  $L_i$  on  $X$ . We then divide the problem into two cases:

Case 1. If there is an  $L_i$  such that  $\text{VB}_S(L_i)$  contains a dense open subset of  $S$ , then  $X \rightarrow S$  is Moishezon (by Proposition 2.7) and we are done.

Case 2. Assume that every such line bundle  $L_i$  has nowhere dense very big locus  $\text{VB}_S(L_i)$ . We claim

$$\text{MO}_S(X) \subset \cup_i \text{VB}_S(L_i) \bigcup \cup_j (e_2(\ell'_j) = 0).$$

If  $s \in \text{MO}_S(X)$ , and  $s \notin \cup_j (e_2(\ell'_j) = 0)$ , we claim that in this case every line bundle on  $X_s$  is numerically equivalent to some  $L_i|_{X_s}$ . For otherwise, there exists a line bundle  $L_s$  on  $X_s$ , with  $c_1(L_s)$  lifting to some  $\ell'_j$ . Since the diagram below commutes, which means that  $\text{ev}_s(e_2(\ell'_j)) = e_2(c_1(L_s)) = 0$  must vanish, this contradicts  $s \notin (e_2(\ell'_j) = 0)$ .

$$\begin{array}{ccc} H^0(S, R^2 g_* \mathbb{Z}) = H^2(S, \mathbb{Z}) & \xrightarrow{e_2} & H^0(X, R^2 g_* \mathcal{O}_X) \\ \simeq \downarrow & & \downarrow \text{ev}_s \\ H^2(X_s, \mathbb{Z}) & \xrightarrow{e_2} & R^2 g_* \mathcal{O}_X(s) \simeq H^0(X_s, \mathcal{O}_{X_s}) \end{array}$$

(Note that the isomorphism  $H^2(X_s, \mathcal{O}_s) \simeq R^2g_*\mathcal{O}_X(s)$  at the point  $s \in \text{MO}_S(X)$  holds since  $R^2g_*\mathcal{O}_X$  is locally free in a neighborhood of  $s \in S$ , using Proposition 3.9 and the Hodge decomposition we proved earlier).

Thus  $X_s$  has a big line bundle (as  $s \in \text{MO}_S(X) \Leftrightarrow L_i|_{X_s}$  is big for some  $i \Leftrightarrow L_i|_{X_s}$  is very big for some  $i$  (and therefore  $s \in \cup_i \text{VB}_S(L_i)$ ). This completes the case when  $R^2g_*\mathcal{O}_X$  is torsion free.  $\square$

We next show that a fiberwise Moishezon morphism is locally Moishezon if the morphism is smooth. Before proving the result, let us give a locally free criterion for the direct image when the fibers satisfy the Du Bois property.

**Theorem 3.9** (Locally freeness criterion for  $R^if_*\mathcal{O}_X$ , [Kol22a, Theorem 24]). Let  $f : X \rightarrow S$  be a smooth, proper morphism of analytic spaces. Assume that  $H^i(X_s, \mathbb{C}) \rightarrow H^i(X_s, \mathcal{O}_{X_s})$  is surjective for every  $i$  for some  $s \in S$ . Then  $R^if_*\mathcal{O}_X$  is locally free in a neighborhood of  $s$  for every  $i$ .

*Proof.* We begin our proof by noting that, by the direct image theorem, it is enough to show the surjectivity of the base change morphism

$$\phi_s^i : R^if_*\mathcal{O}_X(s) \rightarrow H^i(X_s, \mathcal{O}_{X_s}),$$

for every  $i$ . Indeed, the base change theorem shows that the surjectivity of the base change morphisms  $\phi_s^i$  and  $\phi_s^{i-1}$  implies the local freeness of the direct image  $R^if_*(\mathcal{O}_X)$  (see Hartshorne Corollary 12.9).

Next, by the Theorem on Formal Functions, it is enough to prove this when  $S$  is replaced by any Artinian local scheme  $S_n$ , whose closed point is  $s$ .

By Cartan B it is easy to see the vanishing of  $H^p(S_n, R^if_*\mathcal{O}_X) = 0$ ,  $\forall q, \forall i > 0$ , then by the Leray spectral sequence argument we get

$$H^0(S_n, R^if_*\mathcal{O}_X) = H^i(X_n, \mathcal{O}_{X_n}), \quad \text{for } i \geq 0.$$

On the local Artinian base with the closed point  $s$ , we have the following equality:

$$R^if_*\mathcal{O}_X(s) = H^0(S_n, R^if_*\mathcal{O}_X) = H^i(X_n, \mathcal{O}_{X_n}).$$

The base change morphism thus becomes

$$\psi^i : H^i(X_n, \mathcal{O}_{X_n}) \rightarrow H^i(X_s, \mathcal{O}_{X_s}).$$

Let  $\mathbb{C}_{X_n}$  (resp.  $\mathbb{C}_{X_s}$ ) denote the sheaf of locally constant functions on  $X_n$  (resp.  $X_s$ ) and  $j_n : \mathbb{C}_{X_n} \rightarrow \mathcal{O}_{X_n}$  (resp.  $j_s : \mathbb{C}_{X_s} \rightarrow \mathcal{O}_{X_s}$ ) the natural inclusions. We have a commutative diagram

$$\begin{array}{ccc} H^i(X_n, \mathbb{C}_{X_n}) & \xrightarrow{\alpha^i} & H^i(X_s, \mathbb{C}_{X_s}) \\ j_n \downarrow & & \downarrow j'_s \\ H^i(X_n, \mathcal{O}_{X_n}) & \xrightarrow{\psi^i} & H^i(X_s, \mathcal{O}_{X_s}) \end{array}$$

Note that  $\alpha^i$  is an isomorphism since the inclusion  $X_s \hookrightarrow X_n$  is a homeomorphism, and  $j'_s$  is surjective by assumption. Thus  $\psi^i$  is also surjective.  $\square$

Using this we can prove that a smooth fiberwise Moishezon morphism is a locally Moishezon morphism.

**Theorem 3.10** (Fiberwise Moishezon smooth morphism is locally Moishezon, [Kol22a, Corollary 22]). Let  $g : X \rightarrow S$  be a smooth, proper morphism of normal and irreducible analytic spaces whose fibers are Moishezon. Then  $g$  is locally Moishezon.

*Proof.* Since we proved earlier that Moishezon manifolds admit a strong Hodge decomposition, thus

$$H^i(X_s, \mathbb{C}) \rightarrow H^i(X_s, \mathcal{O}_{X_s}),$$

is surjective for every  $i \geq 0$ . The result then follows directly from Theorem 3.9.  $\square$

## 4 The alternating property of the projective locus

In this last section, we will finish the proof of the alternating property of the projective locus. The following Thom-Whitney stratification theorem is useful in the proof.

**Proposition 4.1** (Thom-Whitney stratification theorem, [Kol22b, Lemma 15]). Let  $f : X \rightarrow S$  be a proper morphism of complex analytic spaces. There exist finite Whitney stratifications  $\mathcal{X}$  of  $X$  and  $\mathcal{S} = \{S_l\}_{l \leq d}$  of  $S$  by locally closed subsets  $S_l$  of dimension  $l$ , with  $d = \dim S$ , such that for each connected component  $S$  (a stratum) of  $S_l$ , the following condition holds.

- (a)  $f^{-1}S$  is a topological fibre bundle over  $S$ , a union of connected components of strata of  $\mathcal{X}$ , each mapped submersively to  $S$ ,
- (b) For all  $v \in S$ , there exist an open neighborhood  $U(v)$  in  $S$  and a stratum preserving homeomorphism  $h : f^{-1}(U) \simeq f^{-1}(v) \times U$  s.t.  $f|_U = p_U \circ h$  where  $p_U$  is the projection onto  $U$ .

In particular, there is a dense, Zariski open subset  $S^\circ \subset S$  such that  $g^\circ : X^\circ \rightarrow S^\circ$  is a topologically locally trivial fiber bundle. Moreover, if  $S = \Delta$ , if we shrink the disc then  $f : X^* \rightarrow \Delta^*$  is a topological fiber bundle.

Under this assumption, we can prove that the local system  $R^i g_* \mathbb{Z}_X$  is constructible in the analytic Zariski topology for a proper morphism between complex analytic spaces.

**Corollary 4.2** ([Kol22b, Corollary 16]). Let  $g : X \rightarrow S$  be a proper morphism of complex analytic spaces. Then the sheaves  $R^i g_* \mathbb{Z}_X$  are constructible in the analytic Zariski topology.

When considering the global section of a local system, the following result is helpful.

**Lemma 4.3.** Let  $\mathcal{L}$  be a local system on a complex manifold  $S$ . The global section

$$H^0(S, \mathcal{L}) = L^\rho := \{a \in L \mid \rho(\alpha)(a) = a, \forall \alpha \in \pi_1(S, v)\},$$

where  $L$  is the fiber of the local system at the reference point  $v \in S$ , and  $\rho : \pi_1(S, v) \rightarrow GL(L)$  is the monodromy action. In particular, if the base  $S$  is simply connected, then  $H^0(S, \mathcal{L}) = L$ .



**Proposition 4.4** (The alternating property of projective locus, [Kol22b, Proposition 17]). Let  $g : X \rightarrow S$  be a proper morphism of normal, irreducible analytic spaces. Then there is a dense, Zariski open subset  $S^\circ \subset S$  such that

- (1) either  $X$  is locally projective over  $S^\circ$ ,
- (2) or  $\text{PR}_S(X) \cap S^\circ$  is locally contained in a countable union of Zariski closed, nowhere dense subsets.

If  $g$  is bimeromorphic to a projective morphism, then  $X$  is projective over  $S^\circ$ .

**Remark 4.5.** The locally projective condition is necessary in some situations (that is,  $g : X \rightarrow S$  may not be projective over  $S^\circ$ ). Question to be resolved: where do we use "locally" in the proof? Euclidean topology or Zariski topology?

*Proof.* If we restrict our attention to the main strata  $S^\circ$  of the Whitney stratification, the direct image  $R^2g_*\mathbb{Z}_X$  is locally constant. And further restricting to some Zariski open subset, we can also assume that  $R^2g_*\mathcal{O}_X$  is locally free. By passing to the universal cover, we may also assume that  $R^2g_*\mathbb{Z}_X$  is a constant sheaf. Now consider the sheaf exponential sequence

$$R^1g_*\mathcal{O}_X^* \rightarrow R^2g_*\mathbb{Z}_X \xrightarrow{\partial} R^2g_*\mathcal{O}_X.$$

Let  $\Theta$  be a global section of  $R^2g_*\mathbb{Z}_X$ . By Lemma 4.3, we know that  $\Theta \in H^2(X, \mathbb{Z}) = H^0(S, R^2g_*\mathbb{Z}_X)$ . We decompose the cohomology into two disjoint parts,

$$H^2(X, \mathbb{Z}) = V_1 \sqcup V_2,$$

where

$$V_1 = \{\Theta \in H^2(X, \mathbb{Z}) \mid \partial\Theta \equiv 0\}, \quad V_2 = \{\Theta \in H^2(X, \mathbb{Z}) \mid \partial\Theta \not\equiv 0\}.$$

Since we assume that  $R^2g_*\mathcal{O}_X$  is a vector bundle, the vanishing locus is a Zariski closed nowhere dense subset, which we denote  $H_\Theta = V(\Theta)$  for  $\Theta \in V_2$ .

**Case 1.** Given a point  $s \in \text{PR}_S(X)$ , there exists some ample line bundle  $L_s$  on  $X_s$ , and thus under the exact sequence

$$\text{Pic}(X_s) \rightarrow H^2(X_s, \mathbb{Z}) \xrightarrow{\partial} H^2(X_s, \mathcal{O}_{X_s}),$$

$c_1(L_s) = \Theta_{L_s}$  maps to the zero element,  $\partial(\Theta_{L_s}) = 0$ . Since

$$\text{res}_s : H^2(X, \mathbb{Z}) \xrightarrow{\sim} H^2(X_s, \mathbb{Z}),$$

one can lift the class  $\Theta_{L_s} \in H^2(X_s, \mathbb{Z})$  to a class  $\Theta \in H^2(X, \mathbb{Z})$ .

If  $\partial\Theta$  is identically zero, then it lifts to a line bundle  $L \in \text{Pic}(X)$ , such that  $L|_{X_s} = L_s$ , which is ample, and therefore by Grothendieck's ampleness theorem, we know that the morphism is locally projective around  $s \in \text{PR}_S(X)$ .

**Case 2.** Assume that for all points  $s \in \text{PR}_S(X)$ ,  $\partial\Theta$  is not identically zero. Then  $\partial\Theta = 0$  defines a Zariski closed, nowhere dense subset  $H_\Theta \subset S$ . In this case, we know that

$$\Theta \in V_2,$$

and by the commutative diagram,

we know that  $s \in V(\Theta)$ , and thus

$$PR_S(X) \subset \cup_{\Theta \in V_2} V(\Theta).$$

□

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## 1 Overview

The aim of this note is twofold.

(1) We summarize several projectivity criteria for Moishezon varieties. These include the singular version of Kodaira’s projectivity criterion, the Nakai–Moishezon criterion, Seshadri’s criterion, Kleiman’s ampleness criterion, and a projectivity criterion for Moishezon morphisms as developed in [CH24].

(2) We discuss the projective stratification theorem. The ultimate goal is to complete the proof of the following result.

**Theorem 1.1** ([Kol22, Theorem 2]). Let  $g : X \rightarrow S$  be a proper Moishezon morphism of complex analytic spaces and  $S^* \subset S$  a dense, Zariski open subset such that  $g$  is flat over  $S^*$ . Assume that  $X_0$  is projective for some  $0 \in S$ , and the fibers  $X_s$  have rational singularities for  $s \in S^*$ .

Then there is a Zariski open neighborhood  $0 \in U \subset S$  and a locally closed, Zariski stratification  $U \cap S^* = \cup_i S_i$  such that each  $g|_{X_i} : X_i := g^{-1}(S_i) \rightarrow S_i$  is projective.

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## 2 Projectivity criteria

In this section, we summarize some projectivity criteria related to Moishezon varieties.

### 2.1 Kodaira's projectivity criterion

**Proposition 2.1.** Let  $X$  be a compact Kähler variety with rational singularities such that  $H^2(X, \mathcal{O}_X) = 0$ , then  $X$  is projective.

*Proof.* Take the resolution  $\nu : X' \rightarrow X$ , where  $X'$  is a Kähler manifold. Since  $X$  has rational singularities,  $R^i\nu_*\mathcal{O}_{X'} = 0$  for  $i > 0$ . Thus, by the Leray spectral sequence argument,  $H^2(X, \mathcal{O}_X) = H^2(X', \mathcal{O}_{X'}) = 0$  and therefore by Kodaira's projectivity criterion for smooth manifolds,  $X'$  is projective. And therefore  $X$  is a Kähler Moishezon variety with rational singularities. By the result we proved the first time,  $X$  is a projective variety.  $\square$

### 2.2 Nakai-Moishezon ampleness criteria

**Proposition 2.2** ([Kol90, Theorem 3.11]). Let  $X$  be a proper Moishezon space over  $\mathbb{C}$  and let  $H$  be a line bundle on  $X$ . Then  $H$  is ample on  $X$  if and only if for every irreducible closed subspace  $Z \subset X$ , the intersection product  $H^{\dim(Z)} \cdot Z$  is positive.

### 2.3 Seshadri criterion line bundle version

The Seshadri constant was first introduced by Demailly in the early 90s, when he studied Fujita's conjecture.

**Conjecture 2.3.** Let  $X$  be a smooth projective variety of dimension  $n$ , with  $L$  being ample. Then

- (a)  $K_X + (n+1)L$  is globally generated,
- (b)  $K_X + (n+2)L$  is very ample.

**Definition 2.4.** Given a proper analytic space  $X$  and a line bundle  $L$ , the *Seshadri constant* is defined to be

$$\epsilon(L, x) := \inf_{C \ni x} \frac{L \cdot C}{\text{mult}_x C}.$$

He tried to reduce Fujita's conjecture to the bound control of the Seshadri constant.

**Theorem 2.5** ([Dem92]). Let  $X$  be a smooth projective variety of dimension  $n$  with  $L$  being ample. Then the following hold.

- (a) If  $\epsilon(L, x) > \frac{n}{n+1}$  then  $K_X + (n+1)L$  is globally generated,
- (b) If  $\epsilon(L, x) > \frac{2n}{n+2}$  then  $K_X + (n+2)L$  is very ample.

For readers who want to know more about this, please refer to [Dem92].

**Proposition 2.6** ([Kol22]). Let  $X$  be a proper Moishezon space, and  $D$  a divisor on  $X$  (the same is also true for  $\mathbb{Q}$ ,  $\mathbb{R}$  divisors). Then  $D$  is ample if and only if there exists a positive number  $\varepsilon > 0$  such that

$$\frac{(D \cdot C)}{\text{mult}_x C} \geq \varepsilon,$$

for every point  $x \in X$  and every irreducible curve  $C \subseteq X$  passing through  $x$ .

## 2.4 Seshadri criterion cohomology class version

**Lemma 2.7.** Let  $X$  be a normal compact Moishezon variety. Then the canonical map

$$\Phi : N^1(X) \rightarrow N_1(X)^\vee, \quad [D] \mapsto \lambda_D$$

is an isomorphism. Here we define

$$\lambda_D : N_1(X) \rightarrow \mathbb{R}, \quad [T] \mapsto T \cdot D.$$

**Remark 2.8.** For Fujiki varieties with rational singularities the result is also true:

Let  $X$  be a normal compact Fujiki variety with rational singularities. Then the canonical map

$$\Phi : N^1(X) \rightarrow N_1(X)^\vee, \quad \omega \mapsto \lambda_\omega$$

is an isomorphism. Here we define

$$\lambda_\omega : N_1(X) \rightarrow \mathbb{R}, \quad [T] \mapsto T(\omega).$$

Here

$$N^1(X) := H_{\text{BC}}^{1,1}(X),$$

and  $N_1(X)$  is the vector space of real closed currents of bidimension  $(1, 1)$  modulo the following equivalence relation:  $T_1 \equiv T_2$  if and only if

$$T_1(\eta) = T_2(\eta),$$

for all real closed  $(1, 1)$ -forms  $\eta$  with local potentials.

**Proposition 2.9** ([Kol22]). Let  $X$  be a proper Moishezon space over  $\mathbb{C}$  with rational singularities. Then  $X$  is projective iff there is a cohomology class  $\Theta \in H^2(X, \mathbb{Q})$  and an  $\epsilon > 0$  such that

$$\Theta \cap [C] \geq \epsilon \cdot \text{mult}_p C$$

for every integral curve  $C \subset X$  and every  $p \in C$ .

*Proof.* Note that the cup product induces a  $\mathbb{Q}$ -bilinear form

$$(-) \cap (-) : H^2(X, \mathbb{Q}) \times H_2(X, \mathbb{Q}) \rightarrow \mathbb{Q},$$

which will induce a  $\mathbb{Q}$ -linear functional on  $H_2(X, \mathbb{Q})$ . If  $C \mapsto [C]$  gives an injection  $N_1(X, \mathbb{Q}) \hookrightarrow H_2(X, \mathbb{Q})$ , then we can view  $C \mapsto \Theta \cap [C]$  as a  $\mathbb{Q}$ -linear map

$$\Theta \cap (-) : N_1(X, \mathbb{Q}) \rightarrow \mathbb{Q}.$$

By the previous lemma,  $\Theta \cap (-)$  lies in the dual space  $N^1(X, \mathbb{Q})$ . And line bundles span the dual space of  $N_1(X, \mathbb{Q})$ . So there is a line bundle  $L$  on  $X$  and an  $m > 0$  such that  $\deg(L|_C) = m \cdot \Theta \cap [C]$  for every integral curve  $C \subset X$ . Thus

$$\deg(L|_C) = m \cdot \Theta \cap [C] \geq m\epsilon \cdot \text{mult}_p C,$$

for every integral curve  $C \subset X$  and every  $p \in C$ . Then  $L$  is ample by the line bundle version of the Seshadri criterion. Therefore  $X$  is projective.

Note  $C \mapsto [C]$  gives an injection  $N_1(X, \mathbb{Q}) \hookrightarrow H_2(X(\mathbb{C}), \mathbb{Q})$  if  $X$  has rational singularities, as has been discussed in the first note.  $\square$

## 2.5 Kleiman's ampleness criterion for Moishezon spaces

**Proposition 2.10** ([VP21]). Suppose that  $Y$  is a Moishezon space with  $\mathbb{Q}$ -factorial, log terminal singularities and that  $L$  is a Cartier divisor on  $Y$ . Then  $L$  is ample if and only if  $L$  has positive degree on every irreducible curve on  $Y$  and  $L$  induces a strictly positive function on  $\overline{\text{NE}}(Y)$ .

**Remark 2.11.** It remains open whether the result is still true without the  $\mathbb{Q}$ -factorial KLT assumption.

*Proof.* The proof requires the study of rational curves on Moishezon spaces, we will prove it in the next note.  $\square$

## 3 Approximation of the Chow-Barlet 1-cycle space

In this section, we will introduce the main technical tool: the Chow-Barlet cycle space. We will prove that one can approximate the Chow-Barlet 1-cycle space using countably many families of marked curves, which is crucial for the proof of the result Theorem 6.1.

**Definition 3.1** (Chow functor with  $m$ -marked points, [Kol96, Definition I.3.20]). Let  $X$  be an analytic space over  $S$ . Let

$$\text{Chow}_m(X/S)(Z) = \left\{ \begin{array}{l} \text{Well defined families of nonnegative,} \\ \text{proper, algebraic cycles } \mathcal{C} \text{ of } X \times_S Z/Z, \\ s_1, \dots, s_m : Z \rightarrow X, s_i(z) \in \mathcal{C}_z \text{ for all } z \in Z \end{array} \right\}.$$

We call the data in the bracket the *Chow data with  $m$ -marked points*. We say  $C$  is a *pointed curve* if it is a 1-cycle that has one marked point. And we denote the Barlet-Chow 1-cycle space with 1-marked point by  $\text{Chow}_1^1(X/S)$ .

**Lemma 3.2** (Representative of the Chow functor with marked points). Let  $X \rightarrow S$  be a proper morphism between complex analytic spaces. The relative Chow functor with  $m$ -marked points is representable by a complex analytic space  $\text{Chow}_m(X/S)$ .

*Proof.* Since the proof does not appear in the standard references, for completeness we add a proof here. We claim that the Chow functor with marked points is actually represented by a closed

subspace of the original Chow-Barlet cycle space (we call this closed subspace the incident complex subspace). Let

$$\mathcal{U} \rightarrow \text{Chow}(X/S),$$

be the universal family of the Barlet-Chow cycle space (with  $\mathcal{U} \subset X \times_S \text{Chow}(X/S)$  as closed complex subspace). We then define the  $m$ -fold fiber product to be  $X^{(m)} = \underbrace{X \times_S X \times_S \dots \times_S X}_{m\text{-times}}$ .

Let  $P = \text{Chow}(X/S) \times_S X^{(m)}$ , the incident complex subspace is defined to be

$$\text{Chow}_m(X/S) = I = \{(s, x_1, \dots, x_m) \in P \mid x_i \in \mathcal{U}_s, \text{ for all } i\}.$$

We claim that  $I \subset P$  is a closed complex subspace. Indeed, we have the natural projection

$$p_i : P \rightarrow \text{Chow}(X/S) \times_S X, \quad (c, x_1, \dots, x_m) \mapsto (c, x_i),$$

and it is easy to check that the incidence variety can be represented as

$$I = \bigcap_{i=1}^m p_i^{-1}(\mathcal{U}),$$

since  $\mathcal{U}$  is a closed complex subspace in  $X \times_S \text{Chow}(X/S)$ , and therefore as a finite intersection  $I$  is a closed complex subspace in  $P$ .

We then show that  $I$  is the representative of the Chow functor with marked points, that is

$$\text{Hom}_S(T, I) \simeq \text{Chow}_m(X/S)(T).$$

To see this, we first show that given an  $S$ -morphism  $T \rightarrow I/S$  it will induce a Chow data with marked points over  $S$ . Indeed, since  $I \subset \text{Chow}(X/S) \times_S X^{(m)}$ , the first projection

$$\pi_1 : T \rightarrow I \rightarrow \text{Chow}(X/S),$$

will induce a family over  $T$  via pull back. And the second projection

$$\sigma_i = \pi_{2,i} : T \rightarrow I \xrightarrow{q_i} X,$$

will define the section we want. Conversely, given the Chow data  $(\mathcal{Z}, \sigma_1, \dots, \sigma_m)$  with marked points, it will induce a morphism. To see this, by the representative of the standard Chow functor, we know that there exists a morphism  $\phi_{\mathcal{Z}} : T \rightarrow \text{Chow}(X/S)$  such that  $\mathcal{Z} \rightarrow T$  is the pull back family, with  $m$ -sections  $\sigma_i : T \rightarrow X^{(m)}$ . It is easy to check that the induced morphism actually maps into  $I$ ,

$$\phi_{\mathcal{Z}} \times \sigma_i : T \rightarrow I \subset P.$$

□

The following upper semi-continuity result is needed in the proof.

**Lemma 3.3** (upper semi-continuity of the multiplicities, [BM19, Proposition 4.3.10]). Let  $(X_s)_{s \in S}$  be an analytic family of  $n$ -cycles of a complex space  $M$ . Then the function

$$S \times M \longrightarrow \mathbb{N}, (s, z) \mapsto \text{mult}_z(X_s)$$

is upper semicontinuous in the Zariski topology of  $S \times M$ .

*Proof.* The proof of the lemma is a bit complicated and we omit it here.  $\square$

**Remark 3.4.** In particular, let  $f : X \rightarrow S$  be a proper flat morphism of relative dimension 1, and assume that there is a holomorphic section  $\sigma : S \rightarrow X$ . Then the multiplicity

$$\text{mult} : S \rightarrow \mathbb{Z}, \quad s \mapsto \text{mult}_{\sigma(s)} X_s$$

is Zariski upper-semicontinuous.

*Proof.* Since the fibers  $\{X_s\}$  clearly form an analytic family of cycles in  $X$ , and the section map  $\sigma : S \rightarrow X$  is holomorphic,

$$S \rightarrow S \times X \rightarrow \mathbb{N}, \quad s \mapsto (s, \sigma(s)) \mapsto \text{mult}_{\sigma(s)} X_s,$$

is upper semi-continuous.  $\square$

**Theorem 3.5** (Approximation Chow-Barlet 1-cycle space, [Kol22]). Let  $g : X \rightarrow S$  be a proper morphism of complex analytic spaces that is bimeromorphic to a projective morphism. Fix  $m \in \mathbb{N}$ . Then there are countably many diagrams of complex analytic spaces over  $S$ ,

$$\begin{array}{ccc} C_i & \hookrightarrow & W_i \times_S X \\ w_i \downarrow & \uparrow \sigma_i & \\ & W_i & \end{array}$$

indexed by  $i \in I$ , such that

- (1) the  $w_i : C_i \rightarrow W_i$  are proper, of pure relative dimension 1 and flat over a dense, Zariski open subset  $W_i^\circ \subset W_i$ ,
- (2) the fiber of  $w_i$  over any  $p \in W_i^\circ$  has multiplicity  $m$  at  $\sigma_i(p)$ ,
- (3) the  $W_i$  are irreducible, the structure maps  $\pi_i : W_i \rightarrow S$  are projective, and
- (4) the fibers over all the  $W_i^\circ$  give all irreducible curves that have multiplicity  $m$  at the marked point.

*Proof.* By assumption, there is a bimeromorphic morphism  $r : Y \rightarrow X$  such that  $Y$  is projective over  $S$ .

$$\begin{array}{ccc} Y & \xrightarrow{r} & X \\ & \searrow f & \swarrow g \\ & S & \end{array}$$

By the Lemma, the Barlet-Chow cycle space of curves with marked points on  $Y/S$  exists (denote it  $\text{Chow}_1^1(Y/S)$ ) and its irreducible components  $W_i$  are projective over  $S$ . The universal family

$$\mathcal{C} \rightarrow \text{Chow}_1^1(Y/S),$$

parameterizes all pointed curves on  $Y$  in the fiber direction. Let  $W$  be any irreducible component of  $\text{Chow}_1^1(Y/S)$ . We restrict the universal family on that component to  $C^Y \rightarrow W$ .

We then map back the family of curves on  $Y$ :



$$\begin{array}{ccc}
C^Y & \hookrightarrow & W \times_S Y \\
w_Y \downarrow & \uparrow \sigma_Y & \\
W & & 
\end{array}$$

to a family of curves on  $X$ :

$$\begin{array}{ccc}
C & \hookrightarrow & W \times_S X \\
w \downarrow & \uparrow \sigma & \\
W & & 
\end{array}$$

(Note that the family  $w : C \rightarrow W$  is no longer flat, as curves on the fibers can be contracted by  $Y \rightarrow X$ ).

However, it is still proper flat over some dense Zariski open subset  $W^\circ \subset W$ . Since the family is flat over  $W^\circ$ , by Lemma 3.4, the multiplicity of a fiber  $C_w$  at the section  $s$  is an upper semi-continuous function on  $W^\circ$ . For each  $m \in \mathbb{N}$ , let  $W^m \subset W$  be the closure of the set of points  $p \in W^\circ$  for which  $\text{mult}_{\sigma(p)} C_p = m$ . Since the restriction of a projective morphism over a closed subvariety is still projective,  $W^m \rightarrow S$  is a projective morphism.

We finally go back to the original Moishezon morphism  $g : X \rightarrow S$ . Let  $X^\circ \subset X$  be the largest open set over which  $r : Y \rightarrow X$  is an isomorphism. The above procedure gives all irreducible pointed curves that have nonempty intersection with  $X^\circ$ . Equivalently, all curves with a marked point that are not contained in  $X \setminus X^\circ$ . We can now use dimension induction (note that by the result we proved the first time, the restriction  $X \setminus X^0 \rightarrow S$  is a Moishezon morphism, so that we can repeat the same argument). And we can get countably many families of pointed curves that approximate the Chow-Barlet 1-cycle space with 1-marked point.  $\square$

## 4 Projectivity of very general fibers

We can now prove the following theorem, which is the key step in deducing the main result.

**Theorem 4.1** (Projectivity of very general fibers, [Kol22, Proposition 14]). Let  $g : X \rightarrow S$  be a proper morphism of complex analytic spaces and  $S^* \subset S$  a dense, Zariski open subset such that  $g$  is flat over  $S^*$ . Assume that

- (1)  $X_0$  is projective for some  $0 \in S$ ,
- (2) the fibers  $X_s$  have rational singularities for  $s \in S^*$ , and
- (3)  $g$  is bimeromorphic to a projective morphism  $g^p : X^p \rightarrow S$ .

Then there is a Euclidean open neighborhood  $0 \in U \subset S$  and countably many nowhere dense, closed, analytic subsets  $\{H_j \subset U : j \in J\}$ , such that  $X_s$  is projective for every  $s \in U \setminus \bigcup_j H_j$ .

*Proof.* First choose  $0 \in U \subset S$  such that  $X_U$  retracts to  $X_0$ . Since  $X_0$  is projective, it carries an ample line bundle  $L$ . Let  $\Theta \in H^2(X_U, \mathbb{Q})$  be the pull-back of  $c_1(L)$  to  $X_U$ . Note that  $\Theta$  is a topological cohomology class that is usually not the Chern class of a holomorphic line bundle. Let  $(C_s, p_s)$  be any marked curve on the fiber  $X_s$  for  $0 \neq s \in U$ .

Using Theorem 3.5, we can find countably many families of pointed curves, with projective morphisms  $\pi_i : W_i \rightarrow U$ .

$$\begin{array}{ccc} C_i & \hookrightarrow & W_i \times_S X \\ w_i \downarrow & \uparrow \sigma_i & \\ W_i & & \end{array}$$

Let  $J \subset I$  be the index such that  $H_i := \pi_i(W_i) \subset U$  for  $i \in J$  is nowhere dense in  $U$ . Therefore,  $\pi_i : W_i \rightarrow U$  for  $i \in I \setminus J$  will dominate  $U$ . Since  $\pi_i$  is projective, in particular it implies  $0 \in \pi_i(W_i)$  for  $i \in I \setminus J$ .

Let  $s \in U \setminus \bigcup_{j \in J} H_j$ , then by definition of  $J$ , there is an  $i \in I \setminus J$ , such that the following conditions hold.

- (a)  $(C_s, p_s)$  is one of the fibers of  $w_i$  over  $W_i^\circ$ ,
- (b)  $\text{mult}_{\sigma_i(p)} C_p = m$  for all  $p \in W_i^\circ$ , and
- (c)  $\pi_i : W_i \rightarrow U$  is projective and its image contains  $0, s \in S$  (say  $\pi_i(0) = 0, \pi_i(w) = s$ )

Since  $W_i$  is irreducible, there exists a holomorphic curve  $\tau : \Delta \rightarrow W_i$  connecting the points  $0, w$  (with  $\tau(0) = 0, \tau(1) = w$  and the radius of  $r(\Delta) > 1$ ). We then pull the family back to the disc

$$w : \mathcal{C} \rightarrow \Delta,$$

with section  $\sigma : \Delta \rightarrow \mathcal{C}$ . Note that

$$\text{mult}_{\sigma(t)} \mathcal{C}_t = \text{mult}_{\sigma(1)} \mathcal{C}_1 = \text{mult}_{\sigma_i(s)} C_{is} \text{ for all } t \in \Delta^*,$$

since  $\tau(\Delta^*) \subset W_i^\circ$ . On the other hand, by Lemma 3.4, we have

$$\text{mult}_{\sigma(0)} \mathcal{C}_0 \geq \text{mult}_{\sigma(t)} \mathcal{C}_t = \text{mult}_{\sigma_i(s)} (C_i)_s, \text{ for } t \in \Delta^*.$$

(Here the pull back family  $\mathcal{C} \rightarrow \Delta$  is flat, since the base is a disc and a surjective holomorphic map from a reduced irreducible space to a disc is automatically flat).

Since  $\mathcal{C}_0$  is a 1-cycle on the projective  $X_0$ , and  $\Theta_0 = \Theta|_{X_0}$  is the Chern class of an ample line bundle on  $X_0$ . Thus

$$\Theta \cap [\mathcal{C}_0] \geq \epsilon \cdot \text{mult}_{\sigma(0)} \mathcal{C}_0.$$

by the easy direction of Theorem 2.9, where  $\epsilon$  depends only on  $X_0$  and  $\Theta_0$ .

Since  $\mathcal{C}_0$  and  $\mathcal{C}_1$  lie in the same irreducible component of the Chow-Barlet cycle space, they are algebraically equivalent. Thus the cup product with  $\Theta$  remains the same. Putting these together gives that

$$\boxed{\Theta_s \cap [C_s] = \Theta \cap [\mathcal{C}_1] = \Theta \cap [\mathcal{C}_0] \geq \epsilon \cdot \text{mult}_{p_0} \mathcal{C}_0 \geq \epsilon \cdot \text{mult}_{p_s} C_s}$$

Thus  $X_s$  is projective by the other direction of Theorem 2.9. □

## 5 From locally projective to global projective

**Lemma 5.1** (Trivialization of the monodromy after finite base change). Let  $X$  be a connected complex analytic variety. Let  $\mathcal{L}$  be a local system with finite monodromy defined on  $X$ . Then there exists a finite covering  $\pi : X' \rightarrow X$  such that the pull back local system  $\pi^*\mathcal{L}$  becomes trivial.

*Proof.* Let

$$\rho : \pi_1(X, x_0) \rightarrow \mathrm{GL}_n(L),$$

be the monodromy representation, with  $L$  being the fiber of the local system at the reference point  $x_0 \in X$ . Since the monodromy of  $\mathcal{L}$  is finite,

$$\ker \rho \subset \pi_1(X, x_0),$$

is a finite index normal subgroup. Thus by the Galois correspondence, we can find a finite cover

$$\pi : X' \rightarrow X,$$

such that the fundamental group  $\pi_*(\pi_1(X', x'_0)) = \ker \rho \subset \pi_1(X, x_0)$  with  $\pi(x'_0) = x_0$ . On the other hand, we have the following base change diagram for the monodromy representation.

$$\begin{array}{ccc} \pi_1(X', x'_0) & \xrightarrow{\pi_*} & \pi_1(X, x_0) \\ \rho' \downarrow & & \downarrow \rho \\ \mathrm{GL}(L) & \xrightarrow{=} & \mathrm{GL}(L) \end{array}$$

so that the monodromy of  $\rho' : \pi_1(X', x'_0) \rightarrow \mathrm{GL}(L)$  is clearly trivial.  $\square$

**Lemma 5.2.** Let  $g : X \rightarrow Y/S$  be a proper contraction morphism defined over  $S$ . The induced pull back map on the Néron-Severi group and  $N^1$  space

$$g^* : \mathrm{NS}(Y/S) \rightarrow \mathrm{NS}(X/S), \quad g^* : N^1(Y/S) \rightarrow N^1(X/S),$$

are injective.

**Proposition 5.3.** Assume that  $g : X \rightarrow S$  is a proper Moishezon morphism of normal irreducible analytic spaces. Assume that there exists a dense Zariski open subset  $S^o \subset S$  such that  $X$  is locally projective over  $S^o$ , then it is actually globally projective.

*Proof.* By passing to a Zariski open subset, we may assume that  $R^2g_*\mathcal{O}_X$  is locally free, and  $R^2g_*\mathbb{Z}_X$  is locally constant. Thus by Proposition ?? and Lemma 5.2, after finite base change the Néron-Severi local system becomes a trivial local system, i.e. the locally defined ample line bundle

$$L_i \in \mathcal{NS}(X/S)(U_i) = \mathcal{NS}(X/S)(S),$$

defines a global line bundle.  $\square$

## 6 Kollár's projective stratification theorem

Now we can prove the main theorem of this note.

**Theorem 6.1** (Projective Stratification, [Kol22, Theorem 2]). Let  $g : X \rightarrow S$  be a proper Moishezon morphism of complex analytic spaces and  $S^* \subset S$  a dense, Zariski open subset such that  $g$  is flat over  $S^*$ . Assume that

- (1)  $X_0$  is projective for some  $0 \in S$ ,
- (2) the fibers  $X_s$  have rational singularities for  $s \in S^*$ .

Then there is a Zariski open neighborhood  $0 \in U \subset S$  and a locally closed, Zariski stratification  $U \cap S^* = \cup_i S_i$  such that each

$$g|_{X_i} : X_i := g^{-1}(S_i) \rightarrow S_i \text{ is projective.}$$

*Proof.* By Theorem 4.1, we know that  $\text{PR}_S(X)$  contains the complement of a countable union of Zariski closed, nowhere dense subsets. By the Baire category theorem,  $\text{PR}_S(X)$  is not contained in a countable union of closed, nowhere dense subsets. And by the alternating property of the projective locus that we proved in the previous note, we are in the case that  $g : X \rightarrow S^\circ$  is locally projective over a dense, Zariski open subset  $S^\circ \subset S$ .

Since the morphism is Moishezon, by [Kol22, Complement 18], the morphism  $g : X \rightarrow S$  is globally projective over  $S^\circ$ . And repeating the process on  $S \setminus S^\circ$  gives the stratification of  $g : X \rightarrow S$  into projective morphisms  $g|_{X_i} : X_i = g^{-1}(S_i) \rightarrow S_i$ .  $\square$

## 7 Claudon-Höring's projectivity criterion for Kähler morphisms

In this section, we introduce the following projectivity criterion for Kähler morphisms.

**Theorem 7.1** ([CH24, Theorem 3.1]). Let  $f : X \rightarrow Y$  be a fibration between normal compact Kähler spaces. Assume that  $X$  has strongly  $\mathbb{Q}$ -factorial KLT singularities. Assume one of the following:

- (1) The normal space  $Y$  has klt singularities and the natural map

$$f^* : H^0(Y, \Omega_Y^{[2]}) \longrightarrow H^0(X, \Omega_X^{[2]})$$

is an isomorphism.

- (2) The morphism  $f$  is Moishezon.

Then  $f$  is a projective morphism.

*Proof.* We will discuss this in the next note.  $\square$

Final words: Projectivity of moduli has been systematically studied by Kollár in the 1990's. For readers who want to know more about this direction, please refer to [Kol90].

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